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Experimental Evaluation of Evaporative, Convective and Hybrid Cooling Techniques for Photovoltaic Modules under Outdoor Conditions

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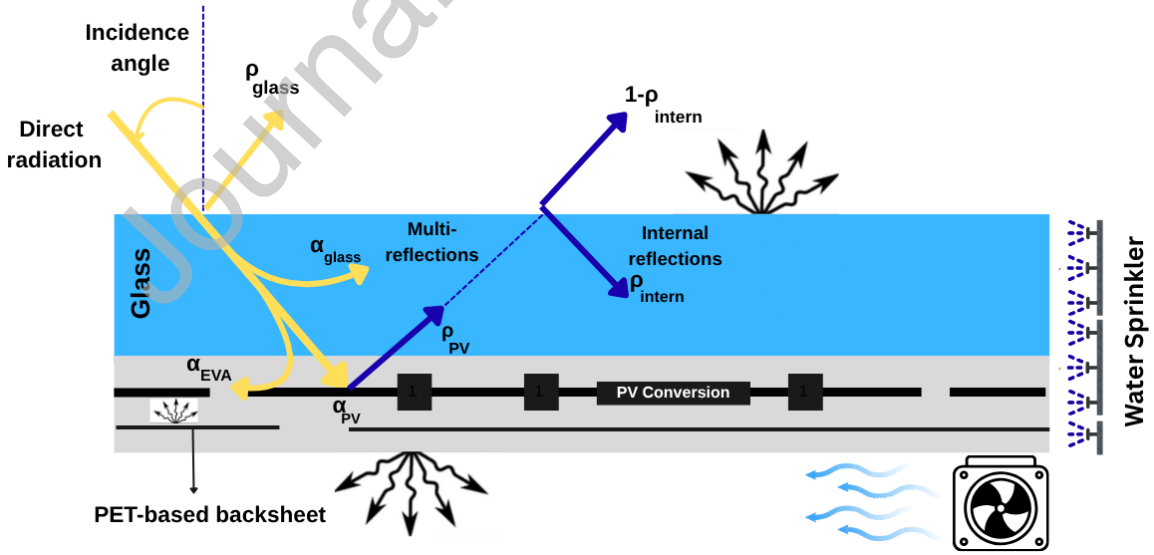
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Abstract

The rapid increase in global energy demand and the environmental impacts of fossil fuels demand integrating efficient solar technologies. Photovoltaic (PV) modules suffer from significant efficiency losses at high temperatures, making cooling strategies essential for improved performance. This study experimentally and comparatively evaluates monocrystalline and polycrystalline PV modules using three cooling techniques, front-surface evaporative water cooling, back-surface forced convection and hybrid cooling configuration under real outdoor winter conditions in Pakistan. Thermal and electrical responses were monitored under varying irradiance, and performance was assessed through power output, efficiency and fill factor. Results demonstrate that water sprinkling yields a peak fill factor of 0.91 ± 0.058 for monocrystalline modules and significantly improves the absolute peak power to 48.0 ± 4.40 W. Hybrid cooling approach produces the most prominent impact, ensuring maximum temperature reductions of 18.5 ± 3.9 °C and 23.5 ± 4.2 °C for monocrystalline and polycrystalline panels, respectively. Furthermore, active cooling sustains peak absolute efficiencies up to $15.98 \pm 1.12\%$ and $14.85 \pm 2.59\%$ in monocrystalline and polycrystalline panels, respectively. Comparative analysis reveals that monocrystalline modules respond more strongly to cooling under greater thermal sensitivity, whereas polycrystalline units demonstrate steadier but smaller improvements. Overall, evaporative water sprinkling and hybrid cooling significantly enhance module reliability, energy yield and long-term stability. Future work should extend analysis across seasonal cycles, incorporate thermal-control methodologies for large-scale PV deployment.

Graphical Abstract



Keywords: Photovoltaic performance; Active thermal management; Evaporative cooling; Forced-air cooling; Hybrid cooling; Energy efficiency

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Nomenclature

Abbreviations		**Greek Symbols**	
CAPEX	Capital Expenditure	α	Absorptivity
DC	Direct Current / Duty Cycle	β	Packing factor
FF	Fill Factor	Δ	Uncertainty / Change in value
GHG	Greenhouse Gas	ϵ	Emissivity
IEA	International Energy Agency	η	Efficiency (%)
IRENA	International Renewable Energy Agency	θ	Angle of incidence/refraction (°)
LCOE	Levelized Cost of Energy	κ	Extinction coefficient
PCM	Phase Change Material	λ	Wavelength
PV	Photovoltaic	μ	Temperature coefficient
PV/T	Photovoltaic/Thermal	ρ	Reflection coefficient
SDG	Sustainable Development Goal	σ	Stefan-Boltzmann constant
UN	United Nations	τ	Transmissivity / Standard deviation
		φ	Porosity
Latin Symbols		**Subscripts**	
A	Panel area (m^2)	a, amb	Ambient
a	Ideality factor	c	Solar cell
C	Cost ($)	$dest$	Destroyed (exergy)
c_p	Specific heat capacity (J/kgK)	eff	Effective
D_{hd}	Hydraulic diameter (m)	f	Fluid (water)
d_{pm}	Particle diameter (m)	g	Glass
E	Power output / Energy	in	Inlet / Input
Ex	Exergy (W)	L	Light-generated
G	Solar irradiance (W/m^2)	mp	Maximum power point
h	Specific enthalpy / Heat transfer coeff.	o	Diode reverse saturation
I	Current (A) / Light Intensity	out	Outlet / Output
k	Thermal conductivity (W/mK)	p	Parallel polarization
L	Thickness (m)	pm	Porous medium
$\dot{m}$	Mass flow rate (kg/s)	ref	Reference conditions
N, n	Refractive index / Number of observations	s	Series / Perpendicular polarization
Nu	Nusselt number	sc	Short circuit
P	Power (W)	sh	Shunt
Pr	Prandtl number	T, ted	Tedlar (back sheet)
R	Resistance (Ω)		
Re	Reynolds number		
S	Specific entropy / Solar radiation		
T	Temperature (°C or K)		
U	Overall heat transfer coefficient (W/m^2K)		
V	Voltage (V) / Air velocity (m/s)		
$\forall, V$	Volume (m^3)		

1. Introduction

Photovoltaic (PV) technology has become one of the most promising renewable energy solutions due to its scalability, declining cost, and wide applicability in distributed and utility-scale power generation. However, the electrical performance of PV modules is strongly affected by operating temperature. Under high solar irradiance, only a fraction of the incident solar energy is converted into electricity, while the remaining part is dissipated as heat, causing a rise in module temperature. This temperature increase reduces open-circuit voltage, lowers power output, and decreases overall conversion efficiency. Therefore, thermal management is essential for maintaining the efficiency, reliability, and long-term energy yield of PV systems, particularly in high-irradiance environments. As the worldwide population is projected to stand at 8 billion figure by 2050, this implies an immense increase in demand for energy about various industries. Economies of superpower countries rely on energy production and consumption at a wide scale. The global attention from fossil fuels is slowly diverting due to depleting reserves and the negative impact on the environment. Conventional fossil fuels such as oil and coal have raised significant environmental concerns as they are responsible for greenhouse gas emissions (GHG), which contribute to global warming and climate change. Thus,

the need of energy generation is consistent with transition from fossil fuels to renewable energy sources to abide by the Sustainable Development Goals (SDGs) of the United Nations (UN) and Paris Agreement. Renewable energy resources including solar, wind, hydro, geothermal, biomass, hydrogen are potential candidates. Among the following, solar energy has emerged as a suitable alternative due to ready availability and abundance [1]. The total capacity of solar photovoltaic (PV) installed globally reached 1418 Gigawatts (GW) by the end of 2023, according to the International Renewable Energy Agency (IRENA), which accounts for 36.7 % of worldwide renewable capability. The magnitude of widespread adoption highlights the potential for solar energy in the years to come [2].

According to International Energy Agency (IEA), an increase in renewable energy resources for power, heat and transport sectors is projected to increase by approximately 60 % between 2024 and 2030. Considering the electricity sector itself, the share of renewable sources is expected to multiply from 30 % in 2023 to 46 % by 2030, with major contribution by solar and wind energy [3]. Solar energy is found in abundance and it can be utilized suing photovoltaic (PV) systems [4] It is by far the most cheapest source of electricity [5]. It is extensively known as photo-electric conversion semiconductors used in PV modules for conversion of direct absorbed solar radiation into electrical power. Silicon cells have conversion efficiencies which match the theoretical maximum values. Achieving full potential for widespread utilization of solar energy and ordinary photovoltaics implies overcoming challenges such as land use, intermittency, storage facilities, and integration with external electrical grid. Provided that solar energy is widely considered as sustainable alternative to conventional energy sources, diverse research project aim to regular reviewing and updates on current studies are crucial [6]. Over the past 10 years, the development in PV output capacity has grown exponentially. Solar energy contributed to 600 TWh of power, or 2.4 % of total electricity. By the end of 2025, it is predicted to have achieved 22 % of the total, with potential to reach 70 %. International Energy Agency Photovoltaic Power Systems Program (IEA PVPS) report issues in 2022 implies that the worldwide PV capacity was 945,7 GW in 2022 [7]. The overall performance of a PV panel is dependent on the stabilization of surface heat with increasing radiation of sun directed at the outer layer of panel. Hence, the widespread use of PV is restricted by the optimal working temperature. High temperatures have adverse effects on overall performance of PV modules. An increase in temperature results in an increase in internal resistance of solar cells, which restricts the flow of current, which consequently leads to decrease in voltage [8, 9]. The overall reduction in voltage and current affects the overall performance and power output of PV system. The combined effect of decrease in current and voltage also affects the power output, resulting in a decrease in overall power output of PV modules. Power output is defined as the product of voltage and current. Since both the parameters decrease as a result of rise in temperature, the output power is decreased significantly. Reduction in power also affects the efficiency and performance of PV system [10, 11]. For typical PV module, efficiency is the effectiveness of a panel in converting sunlight into usable electrical energy. High temperature contributes to a major decrease in efficiency of PV modules [12]. The reduction in current and voltage, combined with power loss, results in decline in the overall performance of PV system. The decrease in efficiency affects the energy yield and performance of the system. The performance of PV module improves with decrease in temperature and vice versa.

For maintaining the optimal performance of PV modules, it is necessary to keep them under a certain temperature range. Through adequate cooling, the surface temperature drops gradually restoring the functionality. Different cooling techniques such as ambient air, water, refrigerant, and phase-change materials (PCM) are widely used [13, 14, 15, 16, 17, 18]. Additionally, nanofluids significantly enhance the efficiency and performance of PV modules and PVT collectors [19]. Literature review reveals application of various cooling techniques aimed at reducing the module temperature and increasing the output power. These techniques can be applied on the top surface, bottom surface, or both surfaces of the panel [20].[21] reported an increase in the PV efficiency by 9 %, using active water cooling where the module temperature plummeted to about 20 %. [22] assessed different cooling techniques for PV setup with the water sprinkling system being the most efficient. [23] investigated the influence of phase change material (PCM) integrated with externals fins on performance of PV panels with various configurations, where an efficiency improvement of 5.16 % and 6.59 % was observed along with temperature reduction of 22.3 °C and 12 °C respectively. [24] observed an improvement in electrical efficiency of PV panels from 10.69 to 12.60 % and efficiency of normalized power output from 61.72 to 71.56 % by using aluminum cans filled with paraffin wax for implementation of thermal management in PV panels. [25] conducted a comprehensive study on the influence of water-cooling techniques in enhancing the panel efficiency, with improvements ranging between 6 % to 82.6 % depending on technique. For dusty environments, water spray cooling system is ideal in maintaining the surface temperature up to a certain level, while keeping the panels clean. Through evaporative cooling, an efficiency increase between 7.6 and 23 % is observed sans pumping power. Using front-surface evaporative water cooling system, [26] observed reduction in temperature of modules by 25 % resulting in 14 % increase in output power. By conducting comparative analysis between front-surface evaporative water cooling and control (no cooling), [27] observed 12.5 % increase in module efficiency, crucial for optimal performance. also studied the influence of front-surface evaporative water cooling on modules, reporting 16.3 % increase in power output and reduction in module temperature to 24 °C. With variable water flow rate, [28] conducted an experimental performance evaluation where he reported enhancement of 25.11 % in PV efficiency using front-surface evaporative water cooling system. [29] investigated the influence of spray angle in enhancing the PV performance, where the flow angle was varied from 15°to 50°. It was observed that by decreasing the spray angle to 15°resulted in enhancement in electrical efficiency by 19.78 % and decrement in average PV panel temperature from non-cooled 64 °C to 24°C. [30] conducted a comparative experimental study on improving the PV efficiency through active and passive cooling provided through cotton wicks combined with regular aluminum fins (CWIRAFs) immersed in water and utilizing absorbing plate with copper tubes

attached to the rear side of the PV module respectively. Through both cooling techniques, a reduction in temperature by 31.4 % and 20.8 %, increase in power by 66.6 % and 56.7 %, and enhancement in electrical efficiency by 8.6 % and 7.9 % was observed. Using circulating water from backside of PV module aided in removing excess heat, thus improving the thermal and overall efficiency by 26.3 % and 34.2 % respectively. [31] reported an improvement in PV efficiency between 3.5 % and 6.2 % and 7 % to 9.3 % for PV module coupled with forced air convection cooling and direct evaporative cooling. [32] investigated the improvement in PV module efficiency using various cooling techniques, where temperature drop varied between 20.5 % to 35.4 % on using water sprinkling system, 6 % to 26 % for air cooled system, with hybrid cooling method being the most efficient system, by exhibiting a temperature drop of 40 %. [33] experimented with mist nozzle assembly, where they observed maximum efficiency gains of 7 % and 9.2 % using natural circulation and forced circulated mist respectively. [34] developed a smart PV cooling and cleaning system controlled through Arduino microcontroller, where he observed an improvement in productive capacity of PV panels by 30.29 %. [35] developed a numerical model by replicating PV system operating under peak solar irradiance conditions (900 W/m^2), coupled with air cooling, front-surface evaporative water cooling and porous media cooling. From simulations, it was inferred that water and porous media cooling were the most efficient techniques, with an improvement in net output gain of 30.9 % and 26.3 % respectively.

Although previous studies have demonstrated the benefits of photovoltaic cooling, direct experimental comparisons of various cooling techniques across monocrystalline and polycrystalline modules under identical outdoor conditions remain limited. To address this gap, the present study experimentally evaluates three active cooling strategies, namely front-surface evaporative water cooling, rear-surface forced air-cooling and hybrid cooling, alongside uncooled reference modules under similar outdoor conditions. The principal novelty lies in the simultaneous comparative assessment of both PV technologies under common experimental environment. This allows for consistent evaluation of their thermal and electrical responses without relying on numerical predictions or comparisons across different testing conditions. The findings have practical relevance to rooftop photovoltaic installations, distributed solar-power systems and the thermal management of PV modules operating under warm and high-irradiance conditions.They can assist researchers in photovoltaic thermal management, renewable-energy engineers and system designers in evaluating cooling strategies and identifying configurations allowing further optimization. The experimental results also provide form basis for subsequent investigations into seasonal performance, auxiliary power consumption, water requirements and economic feasibility of cooling-assisted PV systems.

2. Methodology

2.1. Experimental Setup

The experimental investigation was conducted on the rooftop of the main academic block at COMSATS University Islamabad, Sahiwal Campus, Pakistan (30.63 °N, 73.13 °E), where unobstructed solar exposure ensured consistent irradiance conditions throughout the entire testing period. The comparison between monocrystalline and polycrystalline photovoltaic modules was performed using simultaneous outdoor measurements rather than measurements conducted on different days. A total of six photovoltaic modules, namely three monocrystalline and three polycrystalline modules were installed on the same test frame and operated concurrently under the same solar irradiance, ambient temperature, wind conditions, and measurement schedule. Within each photovoltaic technology, one module served as the uncooled reference, one was subjected to rear-surface forced-air cooling, and one was subjected to front-surface water cooling; the hybrid configuration was subsequently applied to the designated modules during controlled test intervals. Consequently, measurements for the two photovoltaic technologies were obtained under temporally matched environmental conditions, allowing their thermal and electrical responses to be compared under equivalent outdoor conditions. Table 1 provides the detailed technical specifications and electrical specifications of polycrystalline and monocrystalline modules used in the experiment.

Table 1: Specifications of the PV module used

Solar PV module parameters	**Monocrystalline**	**Polycrystalline**
Maximum power (P_{mp})	55 W	40 W
Maximum power voltage (V_{mp})	17.6 V	17.6 V
Maximum power current (I_{mp})	3.14 A	2.29 A
Maximum power point efficiency (η_{mp})	18.5%	17.3
Open circuit voltage (V_{oc})	20.65 V	21.8 V
Short circuit current (I_{sc})	3.33 A	2.44 A
Operating Temperature	-40°C–85°C	-40°C–85°C
Nominal Operating Cell Temperature	44.5 $\pm$ 2°C	44.5 $\pm$ 2°C
Temperature co-efficient of short-circuit current (μ_{sc})	3.5 mA/K	3.5 mA/K

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Each set of panels was configured to evaluate different cooling strategies simultaneously, allowing a direct comparison under identical environmental conditions. Specifically, for each PV technology, one panel was operated without cooling (for reference), one was subjected to rear-surface forced-air cooling, and one panel was exposed to water sprinkling. The hybrid cooling configuration was implemented by combining both air and water mechanisms on the respective panels during controlled test intervals. All panels were securely mounted on a rack oriented southwards with an optimal inclination angle of 42°, selected in the basis of site latitude and optimized for winter conditions. This uniform placement ensured that all the panels experience the same solar irradiance, ambient temperature and wind conditions, thereby eliminating environmental variability as a factor in performance comparison. Figure 1 shows the typical positioning of PV modules

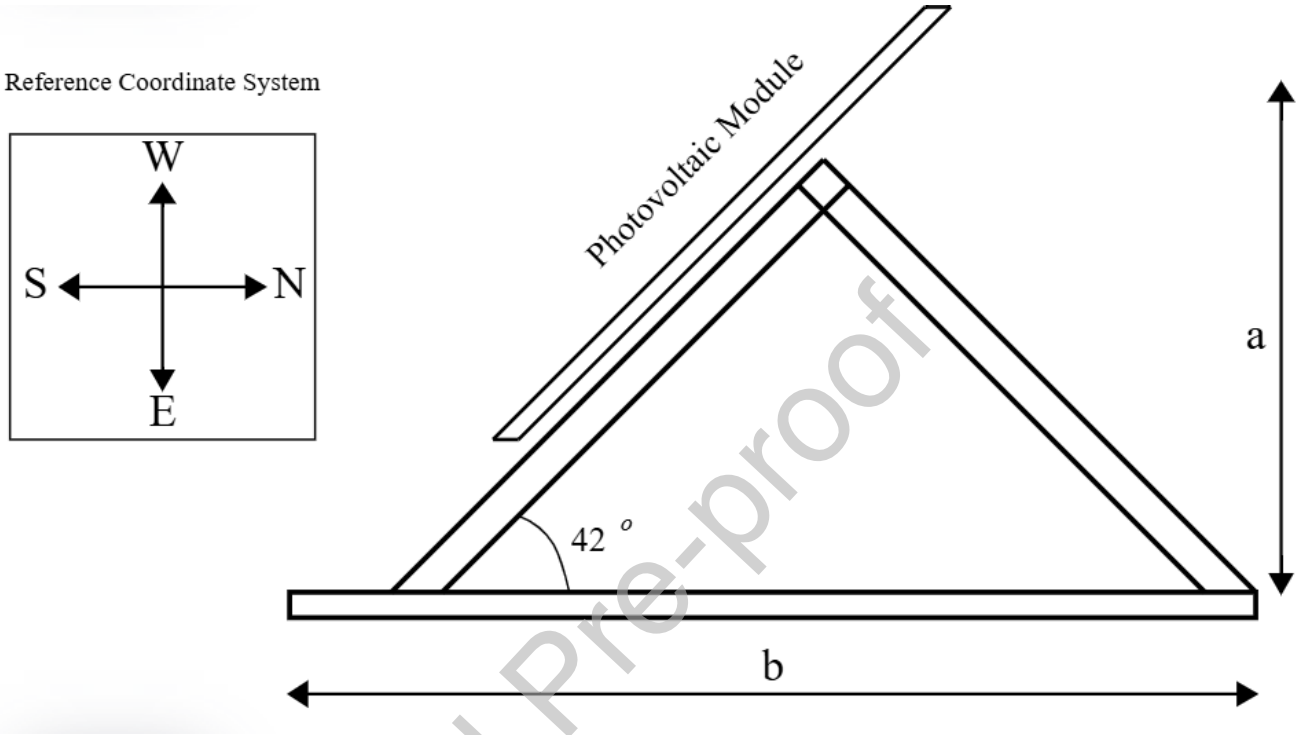


Figure 1: Positioning of PV modules

$$\text{Angle of Inclination} = \text{Latitude} \pm 15^{\circ}(\text{Depending on season}) \tag{1}$$

The cooling configurations were designed to target different heat transfer mechanisms. In the forced rear-surface forced-air cooling setup, four DC fans were installed at the rear of the panels to enhance convective heat dissipation by continuously removing accumulated hot air from the backside. For evaporative cooling, water was uniformly sprayed on to the front surface of the panels using DC pump operating at a controlled flow rate of 9.6 L/min with duty cycle (DC) of 0.2, was aimed at minimizing parasitic power consumption while keeping the glass wet enough to sustain continuous evaporative cooling with nozzles. A series of water spraying nozzles, oriented at 90°, positioned approximately 10 cm above the surface to ensure even distribution and prevent localized overheating. For enhanced convective heat transfer from the rear of modules, 4 brushless DC fans were symmetrically positioned along the backside of panel, with air velocities of 14.4 m/s and 5.7 m/s at inlet and outlet respectively, implying quantifiable validation of the forced convection process. The velocity drop proved that air stream successfully interacted with panel's backside, dissipating accumulated thermal energy. The fans were positioned precisely 3 cm from the backside of panel about the upper region of the PV module. This distance was determined during initial calibration to optimize forced convective boundary layer development while preventing aerodynamic choking that may occur if fans are mounted flush against the surface. Additionally, the upward position was aimed at enhancing the forced convection along the back surface, The arrangement was intended to facilitate the removal of heat from the module and aid airflow toward the upper outlet. The airflow uniformity across the rear panel surface was periodically assessed, ensuring all PV cells experienced a comparable convective heat sink without localized stagnation zones. The hybrid cooling configuration integrated both front-surface water sprinkling and rear-side forced convection. Physically, this setup isolated the two cooling mediums to prevent mechanical interference. The water sprinkling system was restricted to front glass surface, while the brushless DC fans operated continuously on the rear backsheet. This dual-surface configuration was utilized to utilize the high latent heat of vaporization on the sun-facing side and sensible heat removal on the shaded side, thereby maximizing thermal gradient across the module depth. Figure 2 shows various schematics for the experimental setup, each highlighting various setup configurations for different cooling

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techniques used. Figure 2 **a** highlights the schematic for water sprinkling setup, Figure 2 **b** highlights the schematic for forced air convection setup and Figure 2 **c** demonstrates the combined cooling setup, amalgamating the evaporative and convective cooling effects.

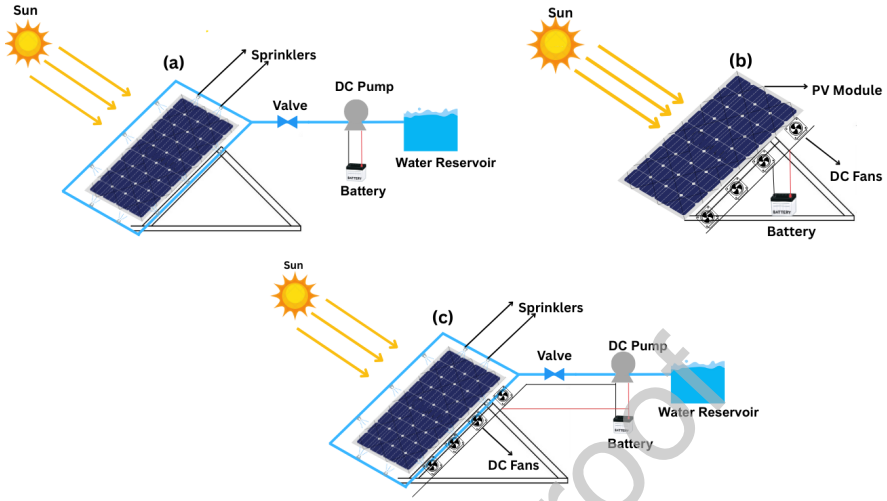


Figure 2: Schematics representation of cooling strategies used for photovoltaic modules (a) water sprinkling system, using DC pump to sprinkle water on panel surface (b) rear-surface forced-air cooling, using DC fans positioned beneath the panels for enhancing convective heat dissipation (c) combined cooling, integrating both water sprinkling and forced air mechanism

2.2. *Experimental Procedure*

The experiments were conducted over a period of 20 consecutive days in November under stable outdoor conditions. The complete dataset was retained for the analysis, however, the results reported in Sections 3.1-3.4 do not display every individual observation from all 20 days to avoid excessive graphical overlap and loss of readability. The selected observations and representative diurnal profiles shown in the figures were used to illustrate the principal thermal and electrical trends, whereas the statistical analysis and reported summary values were based on the complete set of experimental observations described in the methodology. The 20-day duration therefore represents the experimental observation period, while the reduced number of visually distinguishable points in individual figures reflects the presentation strategy rather than a reduction in the experimental duration. To ensure consistency, all panels were operated simultaneously, such that any variations in the solar irradiance or ambient conditions affected all configurations equally. At the beginning of each test cycle, baseline measurements of temperature, voltage, current and irradiance were recorded for all panels without the application of cooling. Subsequently, the designated cooling techniques were activated. The system was allowed a precise five minute stabilization window prior to data acquisition. This duration was established based on empirical pre-testing of the transient thermal response exhibited by PV modules, primarily due to relatively low thermal mass of the front glass and silicon cell layers. Surface temperatures reliably varied within $3-4$ minutes of cooling activation. Therefore, the 5-minute threshold ensures the modules reached a quasi-steady-state thermal equilibrium, allowing the recorded metrics represented sustained cooling performance rather than initial transient thermal shocks.
The experiments were conducted under naturally varying outdoor meteorological conditions. However, wind speed and relative humidity we not systematically recorded at the PV setup location during the experimental investigation. The reported results represent the performance of tested cooling configurations under the prevalent outdoor conditions instead of controlled assessment across specified wind-speed and humidity levels. Six photovoltaic modules, comprising three monocrystalline and three polycrystalline modules, were installed simultaneously within the same outdoor test frame. For each photovoltaic technology, the experimental configurations comprised an uncooled reference condition, rear-surface forced-air cooling, front-surface water sprinkling, and hybrid cooling configuration implemented during controlled test intervals. It is important to mention that the individual module-specific electrical baselines established under a common uncooled condition prior to the cooling trials were not available for the present comparative analysis. Consequently, the experimental design does not fully separate the effects of the cooling treatments from potential differences in the initial electrical characteristics of individual modules. Baseline measurements were first obtained under the uncooled condition, followed by activation of the designated cooling configuration and a five-minute stabilization period before data acquisition. Measurements were taken at specific times of the day, particularly at

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11 : 30 am and 1 : 30 pm, as well as across full diurnal cycles, to capture the variation in performance under different irradiance levels. All observations were replicated in triplicate to ensure reliability and minimize experimental uncertainty. Simultaneous operation of the modules was adopted to minimize differential effects associated with changes in solar irradiance and ambient conditions; however, minor outdoor environmental fluctuations could not be completely eliminated. Key parameters, including surface temperatures, solar irradiance, voltage, current and air velocity were measured using calibrated instruments detailed in Table 2. During each measurement interval, the solar irradiance incident on the module plane, module surface temperature, voltage, and current were recorded under the same outdoor operating condition. Electrical measurements were obtained directly from the photovoltaic modules using the PROVA-210 I–V curve analyzer, which provided the voltage–current characteristics and corresponding maximum-power-point quantities. The instantaneous electrical power was subsequently determined from the measured voltage and current according to $P = VI$, while the conversion efficiency was calculated from the electrical power output relative to the incident solar power, $\eta = P/(GA)$, where G is the measured plane-of-array solar irradiance and A is the active module area. The fill factor was calculated from the maximum-power-point, open-circuit-voltage, and short-circuit-current measurements. Thus, the reported electrical performance parameters were derived from measured electrical and optical quantities rather than estimated from nominal module specifications. The performance indicators such as power output, efficiency and fill factor were subsequently calculated. This systematic and controlled experimental approach ensured that the influence of each cooling technique on PV performance could be accurately isolated and evaluated. The choice of water as effective cooling medium was based on its excellent thermal conductivity, and its high latent heat of vaporization which allows it to absorb significant amounts of solar energy as it evaporates from a liquid into a gas. A stabilization period of five minutes was allowed after the application of each cooling technique to ensure that the system reached thermal equilibrium before measurements were recorded, thereby minimizing transient errors. The implementation of hybrid cooling followed a precise activation sequence to ensure stable data acquisition. Initially, baseline uncooled measurements were recorded. Subsequently, the rear-surface DC fans were activated and allowed to reach steady-state RPM. Immediately following fan stabilization, the front-surface DC water pump was engaged in accordance with its duty cycle. This specific sequence ensures that the continuous forced convective airflow was fully established before the intermittent evaporative cooling phase initiated, preventing transient spikes in localized cell temperatures during stabilization period.

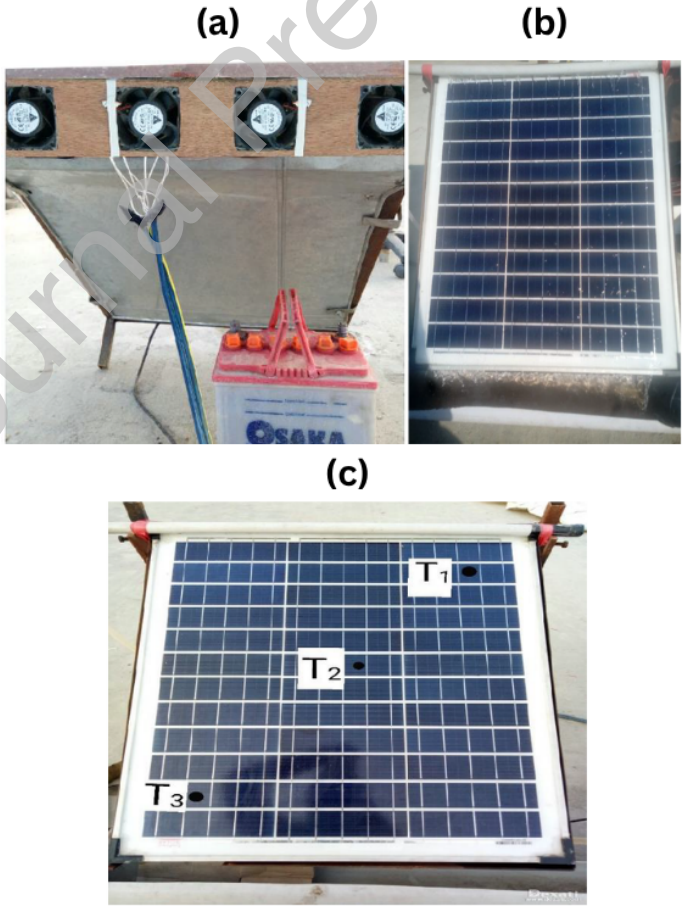


Figure 3: Experimental setup for (a) rear-surface forced-air cooling (b) front-surface evaporative water cooling (c) points for determining panel surface temperature

A comprehensive coupled thermal-electrical-optical mathematical model is provided in Supplementary Data file.

Table 2: Specifications of Equipment Used

Equipment Name	**Specification**
Pyrometer	Temperature Range: -18–550°C (0–1022°F) Resolution: 0.1°C Accuracy: ±1.8% of reading ±1.8°C Response Time: 250ms Spectral Response: 8μm–14μm Laser wavelength: 630nm–670nm
Bevel Protector	Range: 0°–360° Graduation: 5' Accuracy: ±5'
DC Brushless Fans	Model: PFC0612DE Current: 1.68 A Voltage: 12 V(DC) Power Consumption: 20.16 W
Solar Irradiance Meter	Irradiance Measurement Range: 100–1250 W/m^2 Irradiance Resolution: 1 W/m^2 Temperature Measurement Range: -30°C to 125°C Temperature Resolution: 1°C Compass Measurement Range: 0°–360° Compass Resolution: 1° Inclinometer Measurement Range: 0°–90°
PROVA 210 I-V Curve Meter	**DC Voltage Measurement** Range / Resolution / Accuracy 0–10V / 0.001V / ±1% 0–60V / 0.01V / ±1% **DC Current Measurement** Range / Resolution / Accuracy 0.01–10A / 1mA / ±1% 10–12A / 0.01mA / ±1% **DC Current Simulation** Range / Resolution / Accuracy 0.01–10A / 0.1mA / ±1% 0.6–1A / 0.1mA / ±1% Operation Environment: 5°C–50°C Temperature Coefficient: 0.1% of Full Scale Maximum Solar Panel Power Search by AutoScan: 60V, 12A (500W)
Anemometer	Measurement Range: 0.3–45m/s Accuracy: ±3% Temperature Range: 0–45°C Accuracy of Temperature: ±2°C

The measurement instruments listed in Table 2 were used directly during each experimental run. Solar irradiance was measured at the module plane using the calibrated irradiance meter, while module surface temperature was measured using the pyrometer. Voltage and current characteristics were obtained using the PROVA-210 I–V curve analyzer, as stated above. Air velocity was measured at the rear cooling region using the calibrated anemometer. Measurements were performed under the same environmental conditions for the simultaneously operated modules.

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2.3. Net Electrical Power and Energy Accounting

The experiment utilizes 4 DC fans which assist in rear-surface forced air cooling. The corresponding auxiliary power consumption is

$$P_{\text{aux}} = P_{\text{fans}} + P_{\text{pump}} \tag{2}$$

where

$$P_{\text{fans}} = \sum_{i=1}^{4} V_{f,i} I_{f,i} \tag{3}$$

and, for water pump

$$P_{\text{pump}} = V_p I_p \tag{4}$$

The net electrical power of cooled PV module is [36]

$$P_{\text{net,cool}} = P_{\text{PV,cool}} - P_{\text{aux}} \tag{5}$$

where $P_{\text{PV,cool}}$ is the measured PV electrical output under relevant cooling configuration. The cooling improvement relative to the uncooled reference is expressed as

$$\Delta P_{\text{net}} = P_{\text{PV,cool}} - P_{\text{PV,ref}} - P_{\text{aux}} \tag{6}$$

where $P_{\text{PV,ref}}$ is the electrical output of uncooled reference module under comparable conditions. A positive value of P_{net} indicates that the additional electrical output exceeds the auxiliary power requirement.
The net energy gain can be calculated as

$$\Delta E_{\text{net}} = \int_{t_0}^{t_1} [P_{\text{PV, cool}}(t) - P_{\text{PV,ref}}(t) - P_{\text{aux}}(t)]dt \tag{7}$$

For pump rating P_{pump} of 8 W, pump duty cycle D_p of 0.2 and an operation period t_{op} of 7 hours per day, the pump energy can therefore be calculated as

$$E_{\text{pump,day}} = \frac{P_{\text{pump}} D_p t_{op}}{1000} = \frac{8(0.2)(7)}{1000} = 0.0112 \text{ kWh/day} \tag{8}$$

The corresponding annual pump energy is

$$E_{\text{pump,annual}} = (0.0112)(365) = 4.088 \text{ kWh/year} \tag{9}$$

The estimated time-averaged power from pump is

$$P_{\text{pump,avg}} = 8(0.2) = 1.6 \text{ W} \tag{10}$$

For four fans, each consuming 20.16 W and operating for seven hours per day annually, the energy consumed by fans is

$$E_{\text{fans,day}} = \frac{4(20.16)(7)}{1000} = 0.56448 \text{ kWh/day} \tag{11}$$

and

$$E_{\text{fans,annual}} = 0.56448(365) = 206.03 \text{ kWh/year} \tag{12}$$

For hybrid cooling, the combined estimated auxiliary energy would be

$$E_{\text{aux,day}} = 0.0112 + 0.56448 = 0.57568 \text{ kWh/day} \tag{13}$$

The total auxiliary energy is

$$E_{\text{aux,total}} = 4.088 + 206.035 = 210.230 \text{ kWh/year} \tag{14}$$

The net annual energy gain can be interpreted as

$$E_{\text{net,annual}} = E_{\text{gain,annual}} - E_{\text{fans,annual}} - E_{\text{pump,annual}} \tag{15}$$

The total rated capacity for three monocrystalline and three polycrystalline modules, each with rated capacity of 55W and 40 W, is given as

$$P_{\text{rated}} = 3(55) + 3(40) = 285 \text{ W}_p = 0.285 \text{ kW}_p \tag{16}$$

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The base energy E_{base} and energy gain E_{gain} is

$$E_{\text{base}} = (0.285)(1500) = 427.5 \text{ kWh/year} \tag{17}$$

$$E_{\text{gain}} = (0.15)(437.5) = 64.125 \text{ kWh/year} \tag{18}$$

2.4. Uncertainty Analysis

An uncertainty analysis was conducted to quantify the accuracy and reliability of experimental measurements and derived performance parameters, The uncertainty associated with each measured variable depends on the accuracy and resolution of the corresponding instrument, as detailed in Table 2. The surface temperature was measured using a pyrometer with an accuracy of $\pm 1.8\%$ of the reading ± 1.8 °C, while solar irradiance was recorded using a calibrated irradiance meter with resolution of 1 W/m^2. Electrical parameters, including voltage and current were obtained using a PROVA-210 I-V curve analyzer with an accuracy of $\pm 1\%$, and the air velocity was measured using an anemometer with an accuracy of $\pm 3\%$. To improve measurement reliability, all experiments were repeated three times, and the recorded values were averaged. The mean value of any measured quantity X is expressed as

$$\tilde{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{19}$$

where n represents the number of observations, The associated uncertainty due to repeatability is estimated through standard deviation:

$$\sigma_X = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \tilde{X})^2} \tag{20}$$

For the derived parameters, such as power output and efficiency, uncertainty propagation was evaluated using the root-sum-square method, The electrical power is calculated as:

$$P = V \times I \tag{21}$$

and the corresponding uncertainty in power ΔP, is given by

$$\Delta P = \sqrt{(I\Delta V)^2 + (V\Delta I)^2} \tag{22}$$

where ΔV and ΔI are the uncertainties in voltage and current. The efficiency of PV module is expressed as

$$\eta = \frac{P}{GA} \tag{23}$$

where G is the solar irradiance and A is the panel area, The propagated uncertainty in efficiency, $\Delta\eta$ is given by:

$$\Delta\eta = \eta\sqrt{\left(\frac{\Delta P}{P}\right)^2 + \left(\frac{\Delta G}{G}\right)^2} \tag{24}$$

Despite careful experimental design, some sources of uncertainty remain unavoidable. Environmental variations, including fluctuations in solar irradiance, ambient temperature and wind speed during outdoor testing, can influence the measured parameters. Additionally, instrument-related limitations such as sensor response time, alignment errors and potential non-uniformity in water distribution during evaporative cooling can introduce minor deviations. However, by conducting simultaneous measurements under identical conditions and applying statistical averaging, the overall uncertainty was minimized, ensuring that the reported trends and comparative performance analysis remains robust and reliable.

3. Results and Discussion

3.1. Analysis on Effects of Temperature and Irradiance for PV Panels

Figure 4 illustrates irradiance and corresponding panel surface temperature for monocrystalline and polycrystalline PV modules under uncooled, forced air, water sprinkling and combined cooling strategies. The solar irradiance and module surface temperature are presented together to demonstrate the thermal response of the photovoltaic modules under the corresponding incident solar load. The two quantities were not mathematically combined or averaged into a single parameter. Irradiance represents the incident solar input measured at the module plane, whereas temperature represents the corresponding module surface response under each cooling condition. At the selected observation times, the plotted values correspond to the measurements obtained at those specific times rather than to daily averages. For the full diurnal measurements, the profiles represent the sequential observations acquired throughout the measurement period. Module surface temperature was measured directly on the front

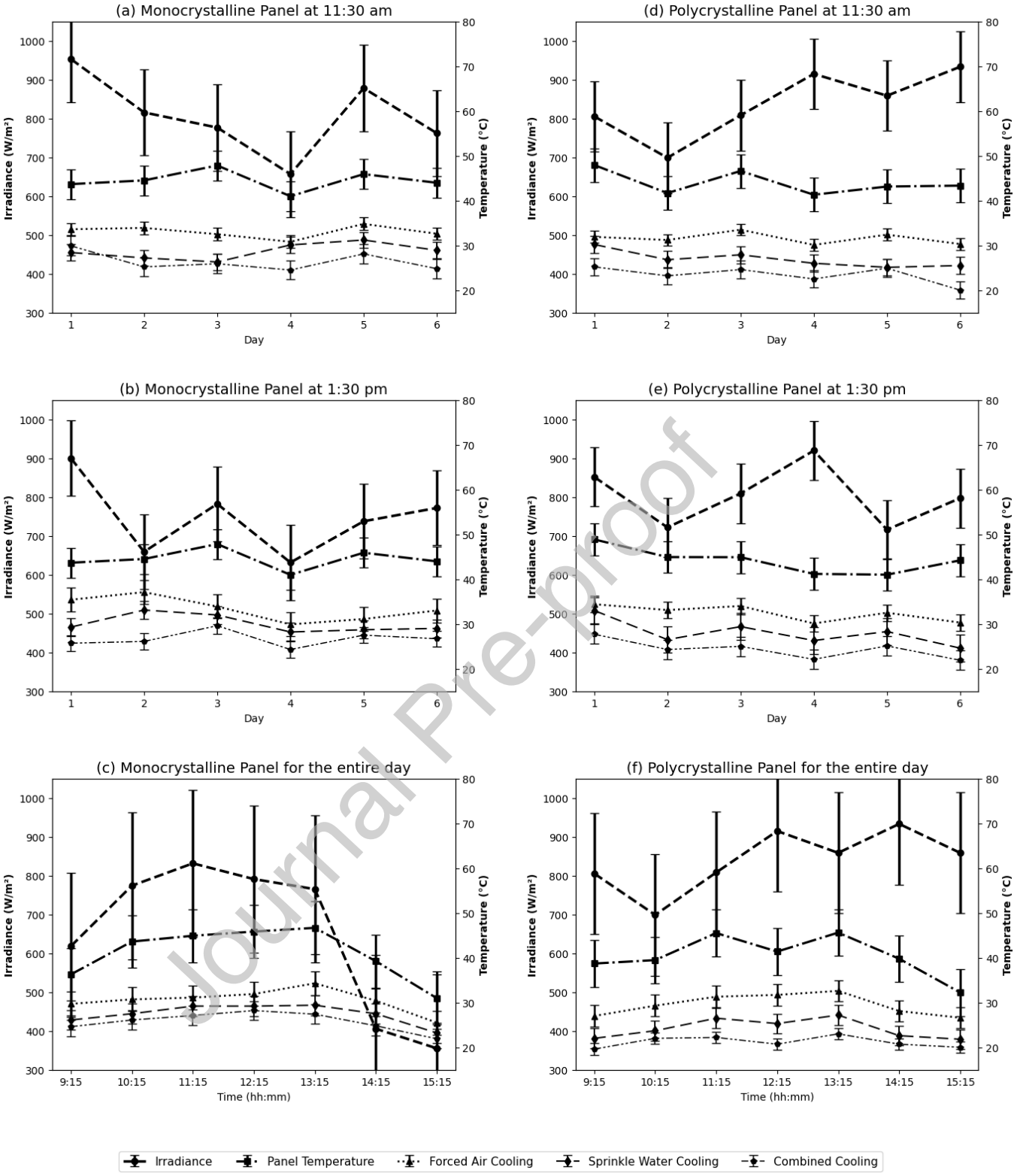
surface of the photovoltaic module using the calibrated pyrometer at representative locations on the module surface, with the same measurement location and procedure maintained for the comparative tests. Increasing irradiance proportionally increases monocrystalline panel temperatures, which are most effectively mitigated through combined cooling (Figures 4 a-c). At $11:30$ am (4 a), with irradiance varying $656-953$ W/m^2, uncooled panels operate between 41.0 °C and 47.8 °C. Forced air achieves moderate $9-13$ °C reduction, bringing the surface temperatures down to $30.8-34.8$ °C. water sprinkling and combined cooling yield major reductions, bringing temperatures to $26.3-31.2$ °C and $24.5-29.9$ °C, respectively. By peak irradiance at $1:30$ pm (Figure 4 b), uncooled monocrystalline temperatures remain high between $41.0-47.8$ °C. Combined cooling provides the maximum temperature drop (achieving $24.3-29.6$ °C), followed closely by water sprinkling $(28.2-33.1$ °C), whereas rear-surface forced-air cooling proves insufficient. Scientifically, water and combined cooling excel because the high latent heat of vaporization extracts substantial thermal energy directly from the front glass. Air cooling is limited by low specific heat capacity of air compared to midday solar thermal loads.

Polycrystalline panels exhibit inherently lower operating temperatures and respond more favorably to air cooling than monocrystalline modules (Figures 4 d-f). At $11:30$ am (Figure 4 d), uncooled polycrystalline modules operate between $41.3-47.9$ °C. Combined cooling significantly reduces temperatures by $20.0-25.2$ °C, while both front-surface evaporative water cooling and forced air achieve comparable reductions up to $25.1-30.2$ °C and $30.1-33.5$ °C, respectively. At $1:30$ pm (Figure 4 e), uncooled panels operate between $41.0-48.9$ °C. Under these afternoon conditions, combined cooling strongly suppresses thermal buildup. bringing temperatures down to $21.9-27.7$ °C. This behavior is due to lower packing density and higher thermal inertia of polycrystalline cells which promote uniform heat distribution [37]. This delayed thermal response allows continuous convective airflow to efficiently remove sensible heat from the backside, making rear-surface forced-air cooling much more viable for polycrystalline modules.

The experimental findings strongly align with the previous studies emphasizing that irradiance increases panel temperature, reducing the open-circuit voltage and efficiency [38]. Monocrystalline modules, due to dense packing and higher absorptivity, suffer greater thermal stresses [39, 40]. They require more aggressive latent-heat cooling to offset voltage and power reductions of $0.3-0.5$ %/°C and $0.4-0.5$ %/°C [41, 42, 43]. Consistent with studies showing evaporative cooling yields $10-20$ °C reductions and up to 31% [30]. The experimental results of this experiment confirm that front-surface evaporative water cooling is the dominant heat extraction mechanism, reducing temperatures by $20-25\%$ [44, 32]. Conversely, polycrystalline modules show narrower temperature variations due to higher thermal inertia [45, 46], validating why forced air remains a highly effective standalone option for this specific module type [30, 37].

Overall, the temperature results confirm that hybrid cooling provided the most effective thermal control for both PV technologies, while the stronger response of monocrystalline modules indicates greater thermal sensitivity.

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Figure 4: Comparison of temperature and irradiance for monocrystalline and polycrystalline PV panels at different times of day. The purple line indicates the panel temperature before application of any cooling technique

3.2. Analysis of Power Production in PV Modules

Figure 5 demonstrates the power output measurements for monocrystalline and polycrystalline modules under uncooled, rear-surface forced-air cooling, front-surface evaporative water cooling and hybrid cooling. The electrical power profiles were generated directly from the voltage and current measurements obtained using the I–V curve analyzer. For each measurement interval, the analyzer recorded the electrical response of the corresponding module, from which the maximum-power-point voltage

and current were identified. The power output was then calculated as $P_{\mathrm{mp}} = V_{\mathrm{mp}} I_{\mathrm{mp}}$. Accordingly, the power profiles shown in the figures represent measured electrical operating conditions rather than values inferred solely from nominal module ratings. Measurements were acquired at the prescribed observation times and during the full diurnal measurement cycles, allowing the effect of cooling to be evaluated both at selected reference times and over the daily operating period. Active cooling significantly mitigates temperature-induced power losses in monocrystalline modules (Figure 5 a-c). At $11:30$ am (Figure 5 a), uncooled panels produced $35.16-43.16$ W. Water sprinkling and combined cooling increase power output to $35.63-48.00$ W (yielding up yo 11.2% maximum relative gain). Combined cooling achieved $41.33-47.80$ W, representing peak relative improvements up to 19.8% over the course of testing days. At $1:30$ pm (Figure 5 b), during peak temperatures and irradiance levels, uncooled power output exhibited wider variance due to severe thermal stress, ranging between $26.15-42.35$ W. Combined cooling effectively restored power to a range of $29.78-46.65$ W. rear-surface forced-air cooling providing moderate yet weaker gains, between $31.45-44.31$ W, primarily due to ineffectiveness of convective airflow in countering thermal buildup. The diurnal pattern (Figure 5 c) shows that under combined cooling, power output peaks at 44.75 W, compared to a maximum of 39.19 W for uncooled module. Combined cooling offers most sustained improvements. While water sprinkling and air cooling improve power output during morning hours, they exhibit varying stability during peak afternoon thermal loads. Scientifically, extracting heat from these densely packed cells suppresses temperature-induced resistance and voltage losses. Forced convective airflow alone cannot counter the massive midday thermal influx, whereas the latent heat of water evaporation aggressively extract thermal energy, ensuring steady electrical current flow.

Polycrystalline modules exhibit inherently lower power outputs but demonstrate greater thermal stability under varying cooling regimes (Figure 5 d-f). At $11:30$ am (Figure 5 d), uncooled outputs ranged between $29.50-36.16$ W. Both forced air and water sprinkling increase power by up to 12.2%, relative to the baseline, raising output to 40.03 W and 39.99 W, respectively. At $1:30$ pm (Figure 5 e), uncooled output varied between $28.35-35.50$ W. Water sprinkling and combined cooling effectively restore power boundaries to $29.35-39.43$ W and $27.84-39.04$ W, respectively. Notably, rear-surface forced-air cooling maintained highly stable gains throughout the cycle ($29.12-39.58$ W). The diurnal profile (Figure 5 f) shows cooled outputs peak between $39.81-40.69$ W compared to the 37.52 W uncooled with distinctly smaller error margins than monocrystalline modules. Since polycrystalline materials possess higher thermal resilience and lower baseline conversion efficiencies, they are less sensitive to severe irradiance fluctuations. This delayed thermal response makes simpler forced air cooling highly viable strategy for maintaining steady power output.

These power outputs corroborate established literature indicating PV efficiency by $0.4-0.5\%$/°C. [47]. Monocrystalline panels, despite having higher baseline efficiencies ($18-22\%$ [48]), suffer greater resistive heating due to dense cell packing [49, 50]. Consequently, they require high capacity water-based cooling, which literature confirms enhances power by $10-20\%$ [51, 6], as opposed to rear-surface forced-air cooling which provides limited heat removal [52, 40]. Excessive convective airflow can prematurely dry the evaporative water film before adequate heat is extracted, highlighting the need to balance airflow and water applications to prevent operational variability.

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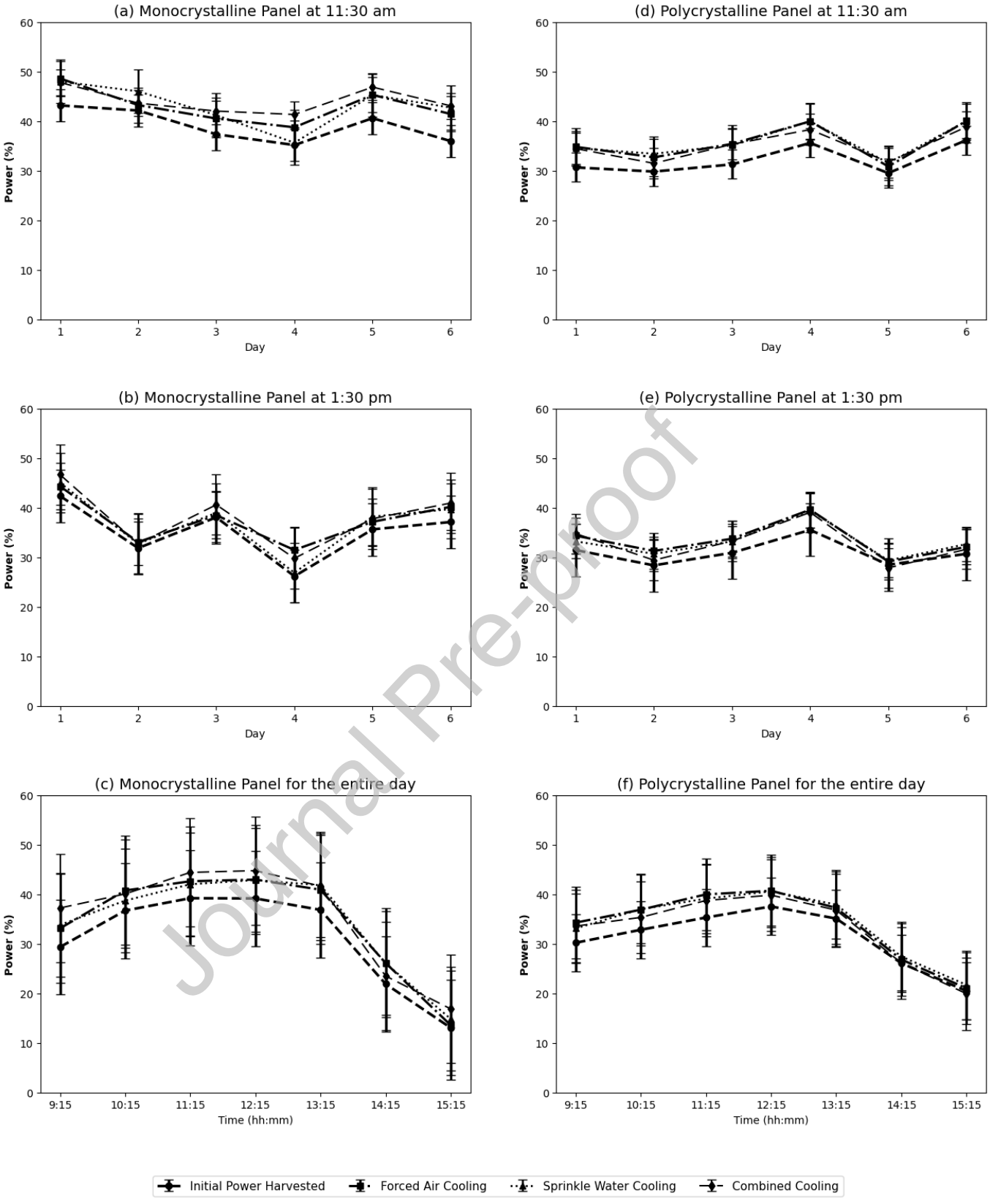


Figure 5: Variation in measured maximum power output of monocrystalline and polycrystalline PV modules at different operating times

3.3. Variation in the Efficiency of PV Modules

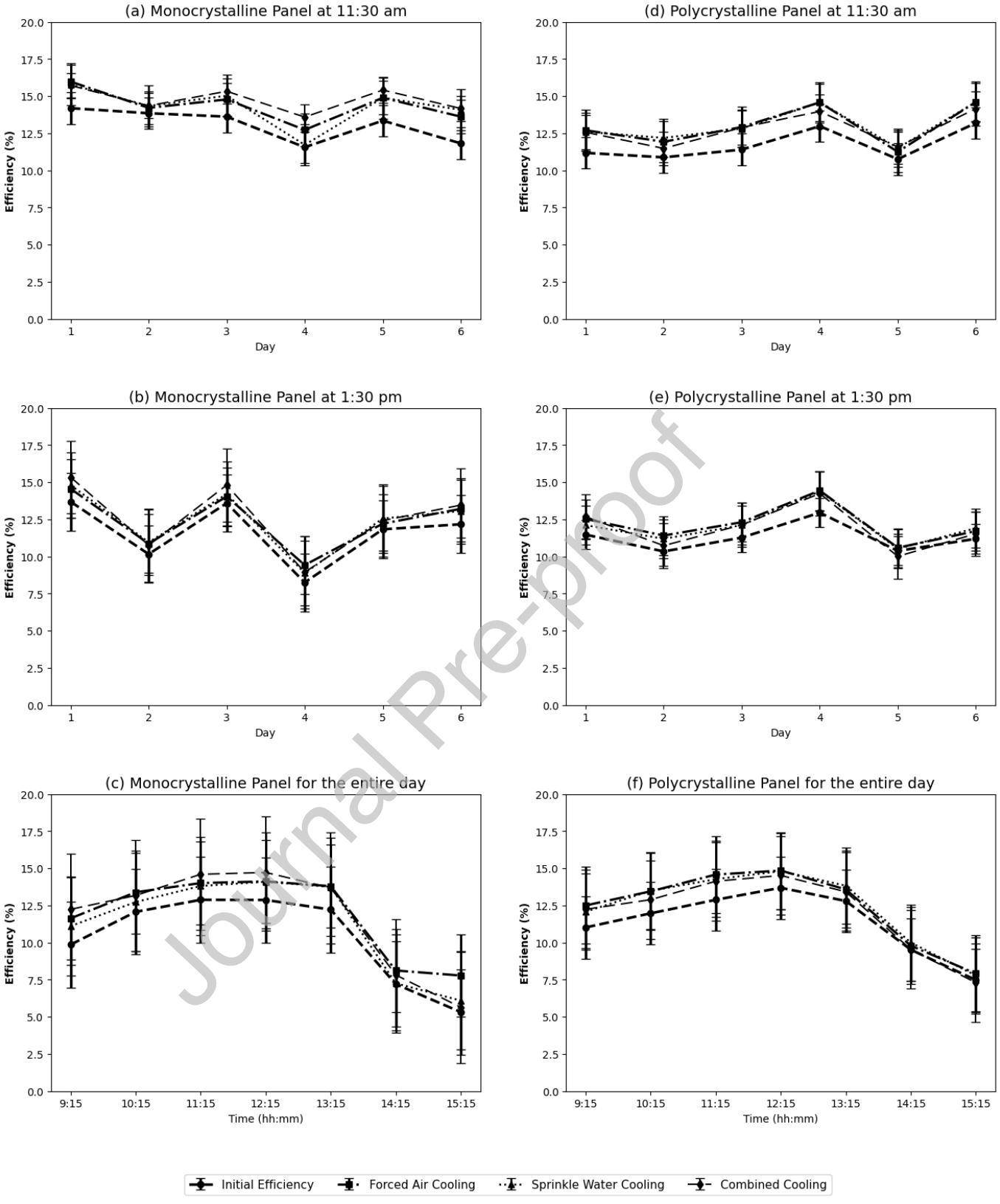
Figure 6 illustrates the conversion efficiency of monocrystalline and polycrystalline PV modules under various cooling strategies across the testing period. Active cooling effectively mitigates sharp efficiency degradation in thermally sensitive monocrystalline panels (Figure 6a-c). At $11:30$ am, uncooled efficiency lies between $11.56 - 14.19\%$. Both combined cooling and

rear-surface forced-air cooling gradually improved efficiency during this timeframe. Air cooling reached peak efficiencies of up to 15.98%, while combined cooling maintained tight, stable operating range of $13.59 - 15.72\%$. Water sprinkling yielded comparable improvements, operating between 11.72% and 15.79%. At $1:30$ pm (Figure 6b), due to severe midday thermal stress, the uncooled efficiency exhibited sharp declines, varying widely between $8.25\% - 13.69$. All active cooling techniques effectively resisted this degradation, with combined cooling restoring the peak efficiency to 15.34%. Across the diurnal cycle (Figure 6c), the efficiency of uncooled module drops significantly in the late afternoon, reaching a low of 5.32% at $15:15$ hours. During these late afternoon hours, rear-surface forced-air cooling maintains an absolute efficiency advantage of $1.2 - 2.5\%$ over the uncooled condition. The combined cooling sustains the highest midday efficiencies, peaking at 14.72%. The phase change of water absorbs massive thermal energy from the glass surface, while the hybrid system's rear fans ensure steady removal of localized hot, humid air, preventing voltage collapse during maximum solar flux.

Polycrystalline modules exhibit slightly lower baseline efficiencies $11 - 13\%$, but demonstrate superior thermal stability (Figure 6 d-f). At $11:30$am (Figure 6 d), the uncooled baseline efficiency ranged between $10.76 - 13.19\%$. Both forced air and water sprinkling enhanced this performance, raising the absolute efficiency to peaks of 14.61% and 14.59%, respectively. At $1:30$ pm (Figure 6 e), uncooled efficiency varied between $10.34 - 12.95\%$, showing substantially smaller performance dips than monocrystalline modules during peak irradiance. Active cooling elevated these boundaries, with rear-surface forced-air cooling reaching up to 14.44% and combined cooling reaching 14.24%. The full-day profile (Figure 6 f) confirms that while polycrystalline modules also experience a late-afternoon efficiency drop, reaching 7.48% uncooled at $15:15$ hours, they maintain highly stable midday operation between $11.02 - 13.69\%$. During peak irradiance hours, forced air and water sprinkling offer absolute efficiency enhancements of approximately 1.0% and 1.5% over the baseline, respectively. Since the multi-crystal structure inherent to polycrystalline panels reduces thermal diffusivity and slows internal heat buildup, continuous forced convective airflow effectively sweeps away sensible heat before it can disrupt electrical conversion, making rear-surface forced-air cooling reliable standalone strategy for the said module type.

These efficiency trends strongly align with the established literature confirming crystalline silicon modules loose $\sim 0.3 - 0.5\%$ efficiency per °C rise [50], corresponding to negative temperature coefficient of $-0.35 - 0.50\%$ °C [53]. The fundamental scientific mechanism for this degradation is enhanced carrier recombination and reduced bandgap energy, which collectively lower the open-circuit voltage. Monocrystalline modules exhibit sharper declines due to high packing density [54]. Cooling mitigates these losses, as water-based cooling restores $8 - 15\%$ efficiency [51, 6], forced air yields $3 - 5\%$ gains, and hybrid cooling maintains above 10% across diurnal cycles [55, 56]. Advanced PV/T systems offer similar heat recovery benefits [57, 40]. In hot climates with limited convection [52, 21], maximizing energy yields requires aggressive water or hybrid cooling for heat-sensitive monocrystalline modules to prevent sharp midday efficiency drops. Simpler rear-surface forced-air cooling provides an effective, highly stable thermal management solution for resilient polycrystalline modules.

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Figure 6: Comparison of efficiency changes for monocrystalline and polycrystalline PV panels at different times of day

3.4. Fill Factor Behavior

Figure 7 illustrates the fill factor (FF) variation for monocrystalline and polycrystalline PV modules under different cooling strategies. The fill factor measures cell quality and deviation from the ideal V-I characteristics, reflecting how effectively cooling prevents internal current leakage. Active cooling dictates significant FF variations in thermally sensitive monocrystalline mod-

ules, with water-based methods proving vastly superior (Figure 7 a-c). At $11:30$ am (Figure 7 a), uncooled panels maintain an FF of $0.716-0.750$. rear-surface forced-air cooling reduces this FF to $0.686-0.737$, whereas water sprinkling yields substantial improvements, operating consistently between $0.744-0.792$. At peak thermal loading at $1:30$ pm (Figure 7 b), water sprinkling drives the highest enhancement, achieving a maximum FF of 0.91, with FF ranging between $0.758-0.910$. Combined cooling provides stable, intermediate resilience ($0.74-0.84$), while forced air shows wider variance of $0.691-0.771$ compared to the uncooled baseline of $0.720-0.748$. Across the full-day profile Figure 7 c), front-surface evaporative water cooling drives prominent noon-hour gains. Scientifically, uneven forced airflow across the module's rear end created localized temperature gradients, inducing electrical mismatch between adjacent cells and subsequently lowering the FF. Conversely, uniform front-surface water sprinkling ensures all cells operate at consistently low temperatures, minimizing internal mismatch losses prior to thermal penetration.

Polycrystalline modules exhibit a highly stable FF and respond favorable to simpler convective cooling (Figure 7 d-f). At $11:30$ am (Figure 7 d), the uncooled baseline FF $(0.738-0.752)$, demonstrating tight thermal bounds. Unlike monocrystalline panels, rear-surface forced-air cooling provides distinct, stable improvements $(0.756-0.766)$, while water sprinkling achieves a comparable maximum value of 0.76. At 1:30 pm (Figure 7 e), uncooled modules maintain a tight fill factor range $(0.738-0.755)$. All active cooling methods providing modest and consistent increments. Forced air elevates the FF to $0.760-0.770$, and water sprinkling reaches $0.755-0.779$. The diurnal profile (Figure 7 f) confirms minimal performance variability across techniques. This behavior is due to inherent properties of polycrystalline modules, which distribute heat more uniformly. This thermal inertia prevents sharp localized temperature spikes, allowing continuous forced air to sufficiently remove sensible heat without inducing electrical mismatches observed in monocrystalline cells.

These observations strictly align with literature demonstrating that FF decreases with rising temperatures due to increased carrier recombination and resistive losses [58, 49]. Since FF depends on junction temperature and series/shunt resistances [59], higher irradiance induces severe thermal stresses that degrade performance unless countered [60]. Lowering the junction temperature mitigates these losses, with high-capacity water methods consistently outperforming rear-surface forced-air cooling for sensitive modules [51, 21, 40].

As a whole, monocrystalline panels demand water sprinkling or hybrid cooling to maintain ideal electrical characteristics during peak solar flux. In contrast, polycrystalline modules possess strong thermal tolerance, making rear-surface forced-air cooling optimal and low-complexity solution. Combined cooling provides robust performance across both types but requires precise optimization of fan speeds and water flow rates to ensure the critical evaporative cooling phase is not prematurely interrupted.

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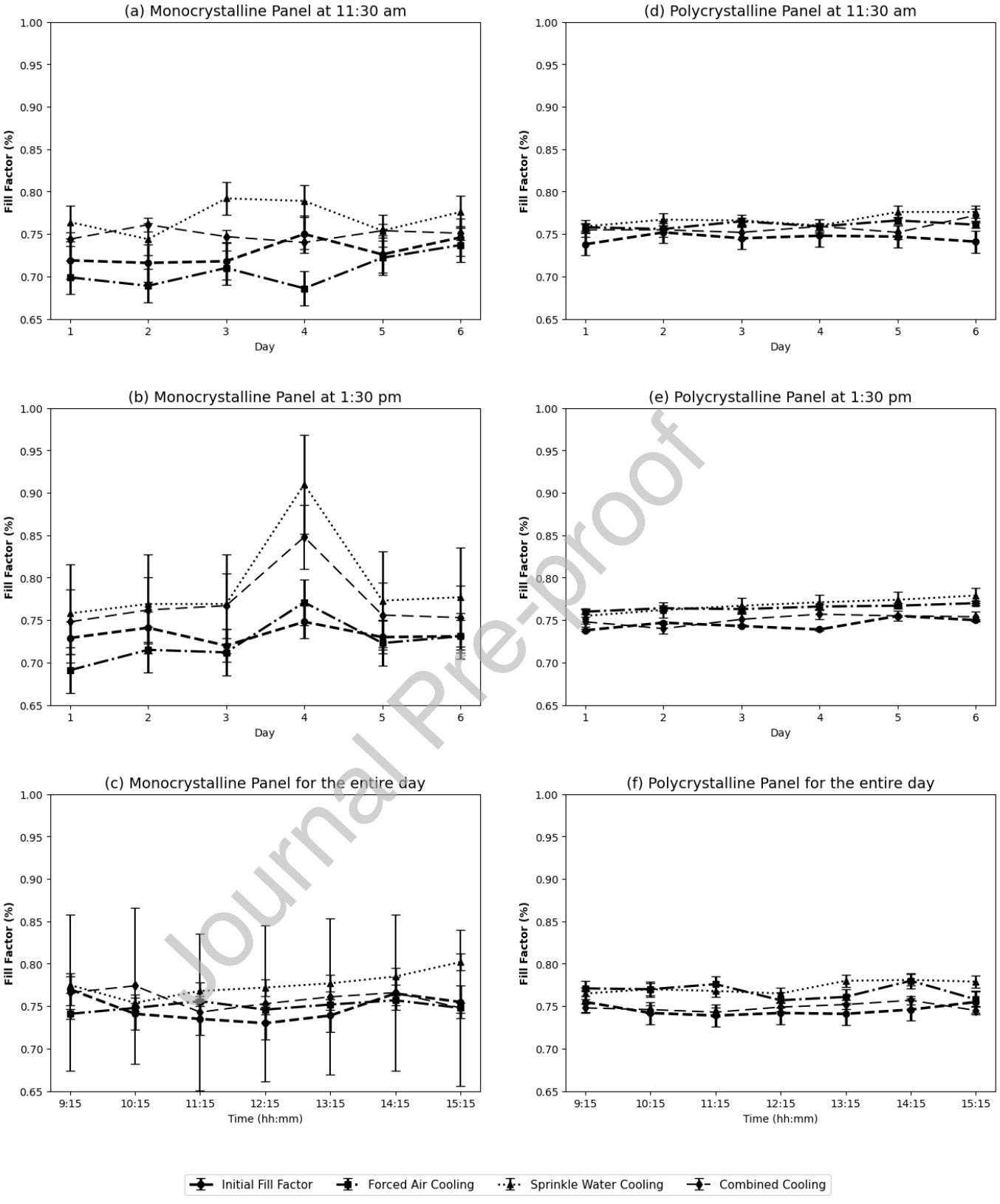


Figure 7: Variation in the Fill factor for Monocrystalline and Polycrystalline PV Modules Across Different Timeframes

3.5. Comparative Analysis of Cooling Techniques on Improvements in PV Performance

A comprehensive analysis regarding the performance of polycrystalline and monocrystalline modules is presented through Subsections 3.1-3.4 where combined cooling technique has emerged as the most suitable cooling method aimed at enhancing the performance of PV modules. The different responses observed for the monocrystalline and polycrystalline modules are associ-

ated with differences in module construction, thermal characteristics, cell arrangement, and temperature sensitivity. However, thermal inertia, cell packing density, and absorptivity were not independently measured in the present investigation. These properties are therefore considered possible contributing mechanisms rather than experimentally established causes of the observed differences. The present interpretation is consequently restricted to the measured thermal and electrical responses under the tested outdoor conditions. Table 3 reports the measured peak values as mean $\pm$ standard deviation, while the corresponding figures identify the photovoltaic technology, cooling configuration, measurement period, and relevant operating conditions. The sample size and statistical significance information have also been clarified to facilitate direct comparison among the tested configurations. Statistical comparisons were performed using one-way analysis of variance (ANOVA) to examine differences among the experimental cooling conditions for each photovoltaic technology and performance indicator. The analysis considered the uncooled condition as the reference and compared it with forced-air, water-sprinkling, and hybrid cooling conditions. Three observations were obtained for each reported experimental condition, consistent with the triplicate measurements described in the experimental procedure. The results are presented as mean $\pm$ standard deviation, with statistical significance evaluated at the 5 % level. Because the present experimental record reports threshold p-values rather than complete ANOVA output, the results are reported as $p < 0.05$ or $p < 0.01$, as applicable, rather than assigning unsupported exact p-values or test statistics. The statistical interpretation was based on the observed experimental replicates, while recognizing that repeated measurements obtained from the same physical modules may not constitute statistically independent experimental units.

Table 3: Comparative analysis of peak performance parameters for monocrystalline and polycrystalline PV modules under different cooling strategies

Cooling Technique	**Monocrystalline**					**Polycrystalline**				
	Max Temp Drop (°C)	**Peak Power (W)**	**Peak Efficiency (%)**	**Peak FF**	***p*-value**	**Max Temp Drop (°C)**	**Peak Power (W)**	**Peak Efficiency (%)**	**Peak FF**	***p*-value**
Forced Air Cooling	13.0 ± 3.6	48.60 ± 3.56	15.98 ± 1.12	0.771 ± 0.027	< 0.05	16.0 ± 4.3	40.69 ± 7.29	14.85 ± 2.59	0.780 ± 0.009	< 0.05
Water Sprinkling	16.0 ± 3.8	48.00 ± 4.40	15.79 ± 1.40	0.910 ± 0.058	< 0.01	18.5 ± 4.7	40.58 ± 6.93	14.81 ± 2.57	0.781 ± 0.007	< 0.01
Hybrid (Air + Water)	18.5 ± 3.9	47.80 ± 2.63	15.72 ± 0.84	0.848 ± 0.038	< 0.01	23.5 ± 4.2	39.81 ± 7.28	14.52 ± 2.64	0.772 ± 0.008	< 0.01

The application of cooling techniques significantly mitigates the thermal stresses developed in monocrystalline and polycrystalline photovoltaic modules. rear-surface forced-air cooling reduced module temperatures by maximum of 13.0 °C and 16.0 °C, respectively. This led to modest performance gains, predominantly more effective for polycrystalline modules owing to inherent thermal stability. The relative lower specific heat capacity of air means that forced air convection alone effectively removes sensible heat at a limited rate. Therefore it struggles against the high thermal loads of densely packed monocrystalline cells but adequately manages the slower heat buildup in polycrystalline cells. Water sprinkling, utilizing evaporative cooling, provided a superior standalone technique for monocrystalline panels, achieving up to 16.0 °C temperature reduction, an 11.2% maximum relative power enhancement, and restoring absolute efficiency to a peak of 15.79%, alongside a maximum fill factor of 0.910. This massive performance improvement is governed by thermodynamics of vaporization, where as the water evaporates, it absorbs excessive latent heat from the glass surface, providing a heat sink in orders of magnitude more effective than dry air. Polycrystalline modules, well-known for being less sensitive to thermal fluctuations, demonstrated steady improvements, including a 12.2% maximum relative power increase and peak absolute efficiency of 14.81%. Hybrid cooling emerged as the most robust cooling technique, for maximizing thermal gradients, achieving a peak temperature reductions of 18.5 °C in monocrystalline panels and 23.5 °C in polycrystalline modules. For monocrystalline panels, hybrid technique yielded a 19.8% maximum relative power gain and sustained peak absolute efficiencies of 15.72%, while for polycrystalline modules, it stabilized performance with a 13.5% power gain and peak efficiency of 14.52%. The advantage of hybrid system lies in its ability to simultaneously utilize latent heat extraction on the front surface while the rear fans actively remove the thermal boundary layer, preventing thermal saturation and maintaining steep temperature gradient between panel and the ambient environment. The performance of evaporative cooling s influenced by ambient humidity and wind conditions. Lower relative humidity increases the potential for evaporation by enhancing the difference between the water vapor pressure at the wet surface and that of the surrounding air. Conversely, higher relative humidity reduces this driving potential and may limit evaporative heat removal. Wind can enhance convective heat and mass transfer at the module surface. However, under spray-based cooling, stronger winds may also deflect droplets, increase spray drift and reduce the uniformity of surface wetting. This means that the cooling performance cannot be associated with the cooling configuration without corresponding meteorological measurements. Overall, water-based methods (as standalone or role in hybrid cooling) outperformed air-based cooling, signifying the critical role of evaporative effects, while the combined cooling ensured consistent long-term improvements by synergizing convective and evaporative mechanisms. It is important to mention that the observations were recorded in the month of November, during cold weather and frequent windy weather. Consequently, the performance improvements reported herein are strictly representative of these specific climate conditions. The baseline thermal dynamics, including natural radiative and convective cooling pathways are inherently connected with seasonal weather profile, While the active, synergistic hybrid cooling effects proved highly beneficial under these conditions, their absolute impact in different climatic zones or under severe thermal loads remain outside the scope of this specific dataset.

3.6. Technoeconomic Assessment: Installation Costs, Energy Consumption, Water Usage and Payback

The technoeconomic assessment presented in this study is intended as an illustrative, scenario-based calculation rather than a comprehensive annual economic feasibility analysis. Since the experimental campaign covered only 20 winter days, the resulting

annualized economic projection is sensitive to seasonal irradiance, ambient temperature, humidity, water losses, electricity tariffs, maintenance requirements, and cooling-system utilization. This preliminary assessment combined the experimentally observed 15 % peak electrical gain with simplified assumptions regarding installation costs, pump energy consumption and year-round operating conditions to provide a baseline economic perspective. Thermal management by front-surface evaporative water cooling and other techniques significantly improves the performance efficiency of PV modules, but introduces capital and operating costs. A review of recent experimental and demonstration studies on water-based and hybrid PV cooling techniques shows that the technoeconomic performance is predominantly governed by installation cost, pump energy consumption, water use strategy and payback time. [61] evaluated pulsed-spray cooling configuration in which detailed CAPEX inputs were incorporated into an LCOE model, reporting installation costs of around \$160/m^2. The results showed that controlling the duty cycle (DC) of spray, especially at DC=0.2, can considerably reduce the water consumption to nearly one-ninth of that required under continuous spraying. Based on the measured pump flow rate of 9.6 L/min operating at DC=0.2, the effective water consumption is 1.92 L/min. Over standard 7-hour operational window, this equates to approximately 806 L of water circulated per panel per day. The practical use of water-spraying cooling depends on the balance between the electrical-performance and the water demand. The use of recirculating water can significantly reduce the requirements for continuous fresh-water input, although water loss through evaporation, droplet drift and system leakage must be replenished. As a result, the quantity of water circulated through the system should not be interpreted as the quantity of freshwater consumed. Intermittent spraying can limit pump operating time. However, the actual water savings depend on the spray distribution, local weather conditions and replenishment requirements. Furthermore, real-world deployment requires strict monitoring of water quality. utilizing hard water with high mineral content can lead to severe scaling and particulate fouling on the front glass. Over time, this scaling degrades optical transmissivity, potentially offsetting the electrical gains achieved through thermal management. The experiment achieved significant LCOE reductions of approximately 46.5 % and 76.3 % for DC=1.0 and DC=0.2, respectively. [20] presented a low-cost cooling kit designed for rooftop 20 kW PV installation, where component costs were minimized by utilizing a simplified design and intermittent flow valve operations. Their analysis implied that intermittent water delivery is essential for reducing the operating water demand, and the overall system can achieve a payback of roughly two years for projected revenues of \$17000 and \$22000 over a 20-year period, as observed in the two case studies performed. In parallel, [62] investigated front-surface evaporative water cooling systems and reported that the installation costs increase with the system size, ranging form approximately 750 euros and 1200 euros for 5kWp and 10 kWp configuration respectively, including the photovoltaic setup installation. Their work highlighted the strong influence of pulsation strategy on pump flow rate and energy expenditure, where pulsations are monitored through flow transmitter, thus providing a practical means of reducing water usage without compromising cooling effectiveness. [63] examined the evaporative cooling using low-cost evaporative pads, made of jute, with reported installation costs of approximately \$23/m^2. their setup required no electrical pump or fan, instead using manual water-level control via 200 L reservoir. Considering the cooling system, a systematic scenario-based assessment for the experimental rig is detailed in Table 4, based on the following equations:

$$\text{Annual additional energy} = \eta_{\text{gain}} \times E_{\text{base}} \tag{25}$$

$$\text{Gross annual benefit} = \text{Annual additional energy} \times P_{\text{elec}} \tag{26}$$

$$\text{Annual operating cost} = E_{\text{pump}} \times P_{\text{elec}} + C_{\text{water}} + C_{\text{maintenance}} \tag{27}$$

$$\text{Net Annual benefit} = \text{Gross annual benefit} - \text{Annual operating cost} \tag{28}$$

$$\tag{29}$$

While the current work emphasized on the thermal and electrical performance, economic feasibility and resource demands are crucial for deployment. Literature review suggests that pulsed-spray or mist system can significantly reduce water consumption, thereby increasing electrical yield and the moderate CAPEX for small-rooftop models has been demonstrated. Pulsed-spray operation reduces water and pumping energy [61] and can considerably reduce the LCOE relative to steady spray. Real-world annual payback periods inherently fluctuate due to seasonal variability, while summer conditions may yield higher absolute temperature reductions, variations in ambient humidity and base module efficiency will alter the net economic benefit.
For stipulated duty cycle of 0.2, which represents 20% pump-on time, the nominal average flow with pump cycle is

$$\bar{Q} = (9.6)(0.2) = 1.92 \text{ L/min} \tag{30}$$

The corresponding gross pumped volume over an operational window t is

$$V_{\text{pumped}} = 1.92(t) = 1.92(420) = 806.4 \text{ L/day} \tag{31}$$

where t is the operation time in minutes. The above estimate represents the volume pumped through the system under the stated operating conditions. The present experimental analysis does not establish measured daily freshwater makeup requirements or a quantitative water balance accounting for evaporation, drainage, leakage and recovered water. Consequently, the freshwater

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Table 4: Scenario-based calculation for cooling system

Observations	Data
Total installed PV capacity	0.285 kWp
Annual baseline annual specific yield	1500 kWh/kWp/year
Calculated baseline annual energy	427.50 kWh/year
Assumed annual electricity gain	15%
Estimated additional annual energy	64.13 kWh/year
Electricity tariff	$0.10/kWh
Gross annual finance benefit	$6.41/year
Pump rated power	8 W
Pump duty cycle	0.2
Assumed daily operating window	7h/day
Calculated annual pump energy	4.09 kWh/year
Number of fans	4
Rated fan power	20.16 W
Calculated annual fan energy	206.04 kWh/year
Total calculated annual auxiliary energy	210.12 kWh/year
Annual auxiliary electricity cost	$21.01/year
Net annual energy balance	146 kWh/year
Net annual finance benefit	$14.60/year
Estimated cooling-kit capital cost	Approximately $170

demand per unit of electrical energy gained cannot be determined reliably from the available measurements. Water availability and operating cost therefore remain important considerations for deployment in water-stressed regions.

3.7. Limitations of the Study

The present study provides a comprehensive experimental evaluation of cooling techniques for photovoltaic modules. However, certain limitations should be acknowledged associated with the study. First, the experiments were conducted over a limited duration of 20 days during the winter season, which may not fully capture the performance variations under extreme summer conditions or across different seasons. Although measurements were collected over 20 consecutive winter days with fixed module inclination of 42°, the experiment represents a single-season campaign at one geographical location. The duration is therefore sufficient for a controlled comparative outdoor assessment but does not establish year-round or seasonal performance. The conclusions are consequently restricted to the tested winter conditions and should not be extrapolated directly to other seasons or climatic regions. Since ambient temperature, solar intensity and wind behavior significantly influence thermal performance, the observed results may vary under different climatic conditions. Under elevated summer irradiance and ambient temperature, the module heat balance, convective heat transfer, evaporative mass transfer and temperature-dependent electrical response may differ substantially. In particular, evaporative cooling depends on ambient humidity, airflow, surface wetting and available vapor-pressure driving force, whereas forced-air cooling depends on the module-to-air temperature difference and the actual airflow conditions.These coupled effects prevent quantitative extrapolation of measured cooling gains to hotter seasons without additional validation. The present study did not include a summer experimental investigation or an independently validated seasonal validation. Consequently, summer-specific cooling performance, evaporation rates and thermal gradients remain undetermined. Wind conditions also influence the practical performance of the front-surface water sprinkling. Increased wind speed can deflect spray droplets away from the module, increase water drift and produce nonuniform surface wetting, hence reducing the cooling effectiveness while increasing water losses. Although the experiments were conducted outdoors under naturally varying conditions, the study did not establish a quantitative relationship between wind speed, spray uniformity and water consumption, nor did it validate a wind-triggered spray-control strategy. A further limitation is the absence of systematic measurements of ambient relative humidity and wind speed at the PV installation.Both parameters strongly influence the evaporative heat and mass transfer and can affect the distribution and retention of sprayed water on the module surface, Their absence prevents a quantitative assessment of how variations in atmospheric moisture and wind conditions contributed to the observed performance differences. Another major limitation concerns the long-term water requirements of front-surface evaporative cooling. Repeated spraying and evaporation of water containing dissolved minerals can potentially produce deposits on the module glass and sprinkler nozzles. Such deposits may reduce optical transmittance, interfere with uniform water distribution and increase maintenance requirements during prolonged outdoor operation. However, mineral deposition, nozzle fouling and changes in front-glass optical transmittance were not quantitatively monitored during the present investigation. Their occurrence and magnitude therefore cannot be established from the available data, and no conclusion regarding long-term scaling resistance or optical durability is drawn. The use of separate PV modules for the different cooling configurations introduces a potential source of experimental uncertainty

because individual modules may differ in their initial electrical characteristics. Since a common uncooled pre-characterization baseline and systematic treatment swapping were not documented, the influence of intrinsic module variability cannot be independently quantifies from the available data. The observed differences should therefore be interpreted as comparisons among the tested module-cooling configurations under the experimental conditions, rather than as fully isolated estimates if the cooling-treatment effect. Additionally, the study was performed at a single geographical location and environmental factors such as humidity, dust accumulation and regional weather patterns were not varied systematically. These factors can significantly influence both the effectiveness of evaporative cooling and the overall thermal behavior of PV modules. Furthermore, while efforts were made to maintain identical operating conditions for all panels, minor fluctuations in irradiance and environmental parameters are inherent to outdoor testing and may introduce small uncertainties in the results. The cooling configurations were also tested at fixed operating settings, and further optimization of parameters such as airflow rate and water distribution could yield improved performance.

Despite these limitations, the study provides reliable comparative insights into the effectiveness of different cooling strategies under real outdoor conditions. Future experiments should incorporate calibrated humidity and wind-speed sensors positioned close to the modules, with measurements synchronized to PV temperature, electrical output and spray orientation. Tests across a broad range of meteorological conditions would allow for development of weather-dependent cooling strategies and a more reliable assessment of technology's performance under various climates. Additionally, studies should characterize the water's dissolved mineral content, inspect nozzle condition and glass deposits at defined intervals. Evaluation of water filtration, treatment and recirculation strategies would also help in determining the maintenance requirements ad practical suitability of evaporative cooling in water-scarce environments. Moreover, studies should also establish module baselines under identical reference conditions, report baseline-normalized electrical changes and consider randomized treatment allocation or crossover testing to reduce module-specific bias.

4. Conclusions

This study presented a comparative, statistically validated experimental evaluation of monocrystalline and polycrystalline PV modules under front-surface evaporative water cooling, rear-surface forced-air cooling and hybrid cooling during winter season.Although the observed thermal responses indicate the potential of active cooling to mitigate module heating, the magnitude and the relative ranking of the resulting benefits under peak summer conditions remain to be established. Based on the rigorous field data and one-way ANOVA testing, primary novel outcomes of the experiment are summarized as follows:

- Hybrid cooling demonstrated the most robust thermal control, achieving highly significant maximum temperature reductions of 18.5 ± 3.9 °C for monocrystalline and 23.5 ± 4.2 °C for polycrystalline modules compared to uncooled baseline.
- Active cooling successfully mitigated temperature-induced losses across both panels. Monocrystalline modules achieved peak power outputs of 48.60 ± 3.56 W and peak absolute efficiencies of $15.98 \pm 1.12\%$. Furthermore, evaporative water sprinkling uniquely optimized the fill factor for monocrystalline panels, reaching an exceptional peak of 0.910 ± 0.058 ($p < 0.01$).
- Monocrystalline modules exhibited pronounced thermal sensitivity, making high-capacity evaporative or hybrid cooling critical for their performance recovery during peak solar flux. In contrast, polycrystalline modules displayed higher thermal inertia. Simpler forced air convection proved to be a reliable standalone strategy for these units, sustaining peak efficiencies of $14.85 \pm 2.59\%$ and a high stable fill factor of 0.780 ± 0.009 ($p < 0.05$).

Integrating advanced thermal management serves as a practical, statistically reliable provider of high energy yield and extended operational lifetimes for PV installations. The scope of these empirical findings is strictly bound to 20-day winter outdoor conditions tested in the study. To establish universally applicable technoeconomic profiles future research must prioritize long-term seasonal validation across diverse, high-temperature climate zones. Subsequent studies should investigate the real-time optimization of water and airflow control parameters, evaluate the determinable effects of panel fouling on evaporative efficiency,and explore the integration of automated, weather-responsive thermal-control methodologies for large-scale PV networks.

4.1. What's Next

The findings of this study have practical implications for improving the thermal management and electrical performance of photovoltaic systems operating under high irradiance and elevated temperature conditions. The observed improvements in efficiency, power output, and fill factor through water sprinkling and hybrid cooling strategies highlight the potential of thermal management as a practical enabler of higher energy yield. While water sprinkling alone proved highly effective in stabilizing panel temperatures, hybrid cooling offered the most consistent and robust performance gains, representing efficiency enhancements of approximately 15.98 ± 1.12 % in monocrystalline and 14.85 ± 2.59 % in polycrystalline modules. This suggests that incorporating advanced cooling methods into conventional PV installations can substantially reduce thermal losses and extend the operational lifetime of modules.

From a scalability perspective, the study's results emphasize the trade-off between cooling effectiveness and resource demands. Water sprinkling, although effective, introduces concerns regarding water consumption, especially in regions already experiencing scarcity. Hybrid cooling systems, while more efficient, require additional infrastructure, synchronization, and potential maintenance efforts that may increase capital and operational costs. Thus, the suitability of each cooling strategy must be evaluated in light of local climatic, economic, and resource conditions. The validated numerical model developed in this work further supports these considerations by providing a predictive framework for assessing PV performance under diverse environmental scenarios, enabling system designers to optimize cooling solutions without relying solely on costly and time-intensive experiments.

At the policy and industry level, the validated effectiveness of hybrid cooling highlights the importance of integrating thermal management strategies into national solar deployment frameworks. Similarly, for residential and commercial rooftop PV installations, where water availability or additional equipment may be limited, low-maintenance forced air or optimized water sprinkling solutions could provide practical performance gains. The modeling framework presented here can serve as a decision-support tool for regulators and investors, helping to quantify the technoeconomic trade-offs of different cooling strategies before implementation. Finally, the broader implication of this research is its contribution to sustainable energy transition goals. By mitigating efficiency losses in PV modules through relatively simple thermal management strategies, the study determines a viable pathway to enhance renewable energy reliability and resilience. This aligns with the targets of the United Nations Sustainable Development Goals (SDGs) and global decarbonization strategies, particularly in countries with abundant solar resources. Moreover, the integration of validated modeling tools with experimental insights provides a strong foundation for developing next-generation PV systems that are not only more efficient but also adaptable to varying climatic and operational challenges.

CRediT Authorship

Muhammad Hasan Iqbal: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Writing - Original Draft. **Muhammad Abdullah Zafar Ghauri:** Software, Formal analysis, Validation, Visualization, Writing - Original Draft, Writing - Review & Editing. **Muhammad Salman Nawaz**:Data curation, Investigation, Writing - Review and editing. **Muhammad Salman Mustafa**:Data curation, Investigation, Writing - Review and editing. **Asad Ali Zaidi:** Writing - Review & Editing. **Muhammad Asad Javed:** Conceptualization, Visualization, Writing - Review & Editing.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] I. O. Harmailil, S. M. Sultan, C. P. Tso, A. Fudholi, M. Mohammad, A. Ibrahim, A review on recent photovoltaic module cooling techniques: Types and assessment methods, 2024. doi:https://doi.org/10.1016/j.rineng.2024.102225.

[2] I. R. E. A. (IRENA), Renewable Capacity Statistics 2024, Technical Report, International Renewable Energy Agency (IRENA), 2024. URL: www.irena.org.

[3] A. O. Ali, A. T. Elgohr, M. H. El-Mahdy, H. M. Zohir, A. Z. Emam, M. G. Mostafa, M. Al-Razgan, H. M. Kasem, M. S. Elhadidy, Advancements in photovoltaic technology: A comprehensive review of recent advances and future prospects, 2025. doi:https://doi.org/10.1016/j.ecmx.2025.100952.

[4] A. Ghosh, Potential of building integrated and attached/applied photovoltaic (bipv/bapv) for adaptive less energy-hungry building's skin: A comprehensive review, 2020. doi:https://doi.org/10.1016/j.jclepro.2020.123343.

Journal Pre-proof

[5] N. Shiradkar, R. Arya, A. Chaubal, K. Deshmukh, P. Ghosh, A. Kottantharayil, S. Kumar, J. Vasi, Recent developments in solar manufacturing in india, Solar Compass 1 (2022) 100009. URL: https://www.sciencedirect.com/science/article/pii/S2772940022000030. doi:https://doi.org/10.1016/j.solcom.2022.100009.

[6] M. Hasanuzzaman, A. B. Malek, M. M. Islam, A. K. Pandey, N. A. Rahim, Global advancement of cooling technologies for pv systems: A review, 2016. doi:https://doi.org/10.1016/j.solener.2016.07.010.

[7] I. E. Agency, Strategic PV Analysis and Outreach PVPS, International Energy Agency, 2022. URL: www.iea-pvps.org.

[8] S. Christopher, M. Vikram, C. Bakli, A. K. Thakur, Y. Ma, Z. Ma, H. Xu, P. M. Cuce, E. Cuce, P. Singh, Renewable energy potential towards attainment of net-zero energy buildings status – a critical review, Journal of Cleaner Production 405 (2023) 136942. URL: https://www.sciencedirect.com/science/article/pii/S0959652623011009. doi:https://doi.org/10.1016/j.jclepro.2023.136942.

[9] P. K. Pathak, D. G. Roy, A. K. Yadav, S. Padmanaban, F. Blaabjerg, B. Khan, A state-of-the-art review on heat extraction methodologies of photovoltaic/thermal system, IEEE Access 11 (2023) 49738–49759. doi:https://doi.org/10.1109/ACCESS.2023.3277728.

[10] A. H. Al-Waeli, K. Sopian, H. A. Kazem, M. T. Chaichan, Photovoltaic/thermal (pv/t) systems: Status and future prospects, 2017. doi:https://doi.org/10.1016/j.rser.2017.03.126.

[11] H. A. Kazem, M. T. Chaichan, A. H. Al-Waeli, K. Sopian, Evaluation of aging and performance of grid-connected photovoltaic system northern oman: Seven years' experimental study, Solar Energy 207 (2020) 1247–1258. doi:https://doi.org/10.1016/j.solener.2020.07.061.

[12] J. K. Tonui, Y. Tripanagnostopoulos, Improved pv/t solar collectors with heat extraction by forced or natural air circulation, Renewable Energy 32 (2007) 623–637. doi:https://doi.org/10.1016/j.renene.2006.03.006.

[13] M. M. Fouad, L. A. Shihata, E. S. I. Morgan, An integrated review of factors influencing the performance of photovoltaic panels, 2017. doi:https://doi.org/10.1016/j.rser.2017.05.141.

[14] M. Hammoud, B. Shokr, A. Assi, J. Hallal, P. Khoury, Effect of dust cleaning on the enhancement of the power generation of a coastal pv-power plant at zahrani lebanon, Solar Energy 184 (2019) 195–201. doi:https://doi.org/10.1016/j.solener.2019.04.005.

[15] J. Chen, G. Pan, J. Ouyang, J. Ma, L. Fu, L. Zhang, Study on impacts of dust accumulation and rainfall on pv power reduction in east china, Energy 194 (2020). doi:https://doi.org/10.1016/j.energy.2020.116915.

[16] X. Liu, S. Yue, L. Lu, J. Li, Investigation of the dust scaling behaviour on solar photovoltaic panels, Journal of Cleaner Production 295 (2021). doi:https://doi.org/10.1016/j.jclepro.2021.126391.

[17] S. M. Shalaby, M. K. Elfakharany, B. M. Moharram, H. F. Abosheiasha, Experimental study on the performance of pv with water cooling, Energy Reports 8 (2022) 957–961. doi:https://doi.org/10.1016/j.egyr.2021.11.155.

[18] H. Yazdani, M. Yaghoubi, Dust deposition effect on photovoltaic modules performance and optimization of cleaning period: A combined experimental–numerical study, Sustainable Energy Technologies and Assessments 51 (2022). doi:https://doi.org/10.1016/j.seta.2021.101946.

[19] M. Lebbi, K. Touafek, L. Boutina, Solar photovoltaic thermal hybrid systems – detailed reviews, Applied Thermal Engineering 279 (2025) 127867. URL: https://www.sciencedirect.com/science/article/pii/S1359431125024597. doi:https://doi.org/10.1016/j.applthermaleng.2025.127867.

[20] A. F. Castanheira, J. F. Fernandes, P. C. Branco, Demonstration project of a cooling system for existing pv power plants in portugal, Applied Energy 211 (2018) 1297–1307. URL: https://www.sciencedirect.com/science/article/pii/S0306261917316859. doi:https://doi.org/10.1016/j.apenergy.2017.11.086.

[21] H. Bahaidarah, A. Subhan, P. Gandhidasan, S. Rehman, Performance evaluation of a pv (photovoltaic) module by back surface water cooling for hot climatic conditions, Energy 59 (2013) 445–453. URL: https://www.sciencedirect.com/science/article/pii/S0360544213006567. doi:https://doi.org/10.1016/j.energy.2013.07.050.

Journal Pre-proof

[22] M. A. Khan, B. Ko, E. Alois Nyari, S. E. Park, H.-J. Kim, Performance evaluation of photovoltaic solar system with different cooling methods and a bi-reflector pv system (brpvs): An experimental study and comparative analysis, Energies 10 (2017). URL: https://www.mdpi.com/1996-1073/10/6/826. doi:https://doi.org/10.3390/en10060826.

[23] R. Kumar, P. Praveen, S. Gupta, J. Saikiran, R. S. Bharj, Performance evaluation of photovoltaic module in-tegrated with phase change material-filled container with external fins for extremely hot climates, Journal of Energy Storage 32 (2020) 101876. URL: https://www.sciencedirect.com/science/article/pii/S2352152X20317138. doi:https://doi.org/10.1016/j.est.2020.101876.

[24] A. D. Tuncer, A. Khanlari, İpek Aytaç, E. Çiftçi, A. Sözen, H. İbrahim Variyenli, Passive thermal management of pho-tovoltaic panel by using phase change material-filled aluminum cans: an experimental study, Heat Transfer Research 53 (2022) 73–86.

[25] M. A. Akrouch, K. Chahine, J. Faraj, F. Hachem, C. Castelain, M. Khaled, Advancements in cooling techniques for enhanced efficiency of solar photovoltaic panels: A detailed comprehensive review and innovative classification, 2025. doi:https://doi.org/10.1016/j.enbenv.2023.11.002.

[26] M. S. Yousef, A. K. A. Rahman, S. Ookawara, Performance investigation of low – concentration photovoltaic systems under hot and arid conditions: Experimental and numerical results, Energy Conversion and Management 128 (2016) 82–94. doi:https://doi.org/10.1016/j.enconman.2016.09.061.

[27] M. Abdolzadeh, M. Ameri, Improving the effectiveness of a photovoltaic water pumping system by spraying water over the front of photovoltaic cells, Renewable Energy 34 (2009) 91–96. doi:https://doi.org/10.1016/j.renene.2008.03.024.

[28] S. A. Polus, R. S. Abdullah, Experimental performance evaluation of tracking photovoltaic system based on variable water flow rate with surface temperature, Energy Sources, Part A: Recovery, Utilization, and Environmental Effects 45 (2023) 5297–5309. URL: https://doi.org/10.1080/15567036.2023.2209533. doi:https://doi.org/10.1080/15567036.2023.2209533. arXiv:https://doi.org/10.1080/15567036.2023.2209533.

[29] M. Nateqi, M. R. Zargarabadi, R. Rafee, Experimental investigations of spray flow rate and angle in enhancing the performance of pv panels by steady and pulsating water spray system, SN Applied Sciences 3 (2021). doi:https://doi.org/10.1007/s42452-021-04169-4.

[30] M. Alktranee, P. Bencs, Experimental comparative study on using different cooling techniques with photovoltaic mod-ules, Journal of Thermal Analysis and Calorimetry 148 (2023) 3805–3817. doi:https://doi.org/10.1007/s10973-022-11940-1.

[31] M. F. Jaffar, A. T. Mohammad, A. Q. Ahmed, W. A. M. Experimental investigation of using the evaporative air cooling technique to enhance the performance of photovoltaic module, International Journal of Low-Carbon Technologies 19 (2024) 1231–1245. URL: https://doi.org/10.1093/ijlct/ctae067. doi:https://doi.org/10.1093/ijlct/ctae067. arXiv:https://academic.oup.com/ijlct/article-pdf/doi/10.1093/ijlct/ctae067/57687439/ctae067.pdf.

[32] I. Al-Masalha, A. S. Alsagagh, O. Badran, N. Alkawaldeh, I. M. Abu-Rahmeh, A. A. Alawin, Improving photovoltaic module efficiency using water sprinklers, air fans, and combined cooling systems, EPJ Photovoltaics 15 (2024). doi:10.1051/epjpv/2024037.

[33] S. A. R. Naqvi, L. Kumar, K. Harijan, A. K. Sleiti, Performance investigation of solar photovoltaic panels using mist nozzles cooling system, Energy Sources, Part A: Recovery, Utilization, and Environmental Effects 46 (2024) 2299–2317. URL: https://doi.org/10.1080/15567036.2024.2305302. doi:https://doi.org/10.1080/15567036.2024.2305302. arXiv:https://doi.org/10.1080/15567036.2024.2305302.

[34] S. S. Hatif, H. K. Latif, A. T. Abdulsadda, Pv panel performance and efficiency improvement using cleaning and cooling techniques, AIP Conference Proceedings 3211 (2025) 040023. URL: https://doi.org/10.1063/5.0274135. doi:https://doi.org/10.1063/5.0274135. arXiv:https://pubs.aip.org/aip/acp/article-pdf/doi/10.1063/5.0274135/20512658/040023_1_5.0274135.pdf.

[35] B. Menacer, N. E. H. Baghdous, S. Narayan, M. Al-lehaibi, L. Osorio, V. Tuninetti, Efficiency enhancement of photovoltaic panels via phase change and porous media cooling methods: Thermal–electrical modeling, Sustainability 17 (2025). URL: https://www.mdpi.com/2071-1050/17/14/6559. doi:https://doi.org/10.3390/su17146559.

[36] M. Lebbi, K. Touafek, A. Benchatti, L. Boutina, A. Khelifa, M. T. Baissi, S. Hassani, Energy performance improvement of a new hybrid pv/t bi-fluid system using active cooling and self-cleaning: Experimental study, Applied Thermal Engineering 182 (2021) 116033. URL: https://www.sciencedirect.com/science/article/pii/S1359431120335146. doi:https://doi.org/10.1016/j.applthermaleng.2020.116033.

[37] T. Ibrahim, M. Abou Akrouch, F. Hachem, M. Ramadan, H. S. Ramadan, M. Khaled, Cooling techniques for enhanced efficiency of photovoltaic panels—comparative analysis with environmental and economic insights, Energies 17 (2024). URL: https://www.mdpi.com/1996-1073/17/3/713. doi:https://doi.org/10.3390/en17030713.

[38] M. A. Koondhar, A. Sahito, Z. Memon, K. Harijan, Performance analysis of monocrystalline and polycrystalline pv modules under outdoor conditions in a desert environment, Renewable Energy 170 (2021) 726–739. doi:https://doi.org/10.1016/j.renene.2021.01.054.

[39] P. Singh, N. M. Ravindra, Dependence of solar cell performance on polycrystalline silicon material properties, Solar Energy Materials and Solar Cells 101 (2012) 36–45. doi:https://doi.org/10.1016/j.solmat.2012.02.018.

[40] A. H. A. Al-Waeli, K. Sopian, H. A. Kazem, M. T. Chaichan, Photovoltaic solar thermal (pv/t) collectors past, present and future: A review, International Journal of Applied Engineering Research 12 (2017) 11273–11285.

[41] K. A. Moharram, M. S. Abd-Elhady, H. A. Kandil, H. El-Sherif, Enhancing the performance of photovoltaic panels by water cooling, Ain Shams Engineering Journal 4 (2013) 869–877. doi:https://doi.org/10.1016/j.asej.2013.03.005.

[42] M. Ijaz, F. Shafiq, M. Asghar, M. Younas, Experimental evaluation of pv module temperature and performance in hot climatic conditions, Energy Conversion and Management 236 (2021) 114014. doi:https://doi.org/10.1016/j.enconman.2021.114014.

[43] A. R. Jordehi, Dust accumulation on photovoltaic panels: A review of the practical implications, detection, and cleaning methods, Renewable and Sustainable Energy Reviews 59 (2016) 1307–1320. doi:https://doi.org/10.1016/j.rser.2016.01.050.

[44] S. Iqbal, S. Afzal, A. U. Mazhar, H. Anjum, A. Diyyan, Effect of water cooling on the energy conversion efficiency of pv cell, American Scientific Research Journal for Engineering, Technology, and Sciences 20 (2016) 122–128. URL: https://asrjetsjournal.org/American_Scientific_Journal/article/view/1680.

[45] M. Benghanem, S. Haddad, A. Alzahrani, A. Mellit, H. Almohamadi, M. Khushaim, M. S. Aida, Evaluation of the performance of polycrystalline and monocrystalline pv technologies in a hot and arid region: An experimental analysis, Sustainability 15 (2023). URL: https://www.mdpi.com/2071-1050/15/20/14831. doi:https://doi.org/10.3390/su152014831.

[46] M. Alktranee, P. Bencs, Effect of evaporative cooling on photovoltaic module performance, Process Integration and Optimization for Sustainability 6 (2022) 921–930. doi:https://doi.org/10.1007/s41660-022-00268-w.

[47] C. Xiao, X. Yu, D. Yang, D. Que, Impact of solar irradiance intensity and temperature on the performance of compensated crystalline silicon solar cells, Solar Energy Materials and Solar Cells 128 (2014) 427–434. URL: https://www.sciencedirect.com/science/article/pii/S0927024814003274. doi:https://doi.org/10.1016/j.solmat.2014.06.018.

[48] M. A. Green, E. D. Dunlop, G. Siefer, M. Yoshita, N. Kopidakis, K. Bothe, X. Hao, Solar cell efficiency tables (version 61), Progress in Photovoltaics: Research and Applications 31 (2023) 3–16. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/pip.3646. doi:https://doi.org/10.1002/pip.3646. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/pip.3646.

[49] E. Radziemska, The effect of temperature on the power drop in crystalline silicon solar cells, Renewable Energy 28 (2003) 1–12. doi:https://doi.org/10.1016/S0960-1481(02)00015-0.

[50] E. Skoplaki, J. A. Palyvos, On the temperature dependence of photovoltaic module electrical performance: A review of efficiency/power correlations, Solar Energy 83 (2009) 614–624. doi:https://doi.org/10.1016/j.solener.2008.10.008.

[51] M. Chandrasekar, T. Senthilkumar, S. Suresh, S. Harish, Experimental investigation on photovoltaic-thermal (pv/t) water based cooling system, International Journal of Electrical Power & Energy Systems 47 (2013) 479–485. doi:https://doi.org/10.1016/j.ijepes.2012.11.034.

Journal Pre-proof

[52] R. Kumar, M. A. Rosen, A critical review of photovoltaic–thermal (pv/t) hybrid solar systems, Applied Energy 88 (2011) 3483–3499. doi:https://doi.org/10.1016/j.apenergy.2011.04.044.

[53] N. Kumari, S. K. Singh, S. Kumar, V. K. Jadoun, Performance analysis of partially shaded high-efficiency mono perc/mono crystalline pv module under indoor and environmental conditions, Scientific Reports 14 (2024). doi:https://doi.org/10.1038/s41598-024-72502-z.

[54] S. Dubey, J. N. Sarvaiya, B. Seshadri, Temperature dependent photovoltaic (pv) efficiency and its effect on pv production in the world – a review, Energy Procedia 33 (2013) 311–321. doi:https://doi.org/10.1016/j.egypro.2013.05.072.

[55] A. E. Kabeel, M. Abdelgaied, Performance enhancement of solar pv modules by using phase change materials (pcm) and nano-fluid cooling: A review, Renewable and Sustainable Energy Reviews 45 (2016) 83–94. doi:https://doi.org/10.1016/j.rser.2015.11.047.

[56] H. A. Hussien, H. A. Hasan, K. Sopian, Performance improvement of photovoltaic panels using hybrid cooling techniques: A review, Renewable and Sustainable Energy Reviews 125 (2020) 109776. doi:https://doi.org/10.1016/j.rser.2020.109776.

[57] T. T. Chow, A review on photovoltaic/thermal hybrid solar technology, Applied Energy 87 (2010) 365–379. doi:https://doi.org/10.1016/j.apenergy.2009.06.037.

[58] S. Amar, M. Bahich, Y. Bentahar, M. Afifi, E. Barj, A study of the temperature influence on different parameters of mono-crystalline silicon photovoltaic module, Journal of Power and Energy Engineering 09 (2021) 29–42. doi:https://doi.org/10.4236/jpee.2021.96003.

[59] M. A. Green, Solar cell fill factors: General graph and empirical expressions, Solid-State Electronics 24 (1982) 788–789. doi:https://doi.org/10.1016/0038-1101(81)90162-9.

[60] M. Wolf, H. Rauschenbach, Series resistance effects on solar cell measurements, Advanced Energy Conversion 3 (1963) 455–479. doi:https://doi.org/10.1016/0365-1789(63)90063-8.

[61] A. Hadipour, M. Rajabi Zargarabadi, S. Rashidi, An efficient pulsed- spray water cooling system for photovoltaic panels: Experimental study and cost analysis, Renewable Energy 164 (2021) 867–875. URL: https://www.sciencedirect.com/science/article/pii/S0960148120314294. doi:https://doi.org/10.1016/j.renene.2020.09.021.

[62] K. Sornek, W. Goryl, R. Figaj, G. Dąbrowska, J. Brezdeń, Development and tests of the water cooling system dedicated to photovoltaic panels, Energies 15 (2022). URL: https://www.mdpi.com/1996-1073/15/16/5884. doi:https://doi.org/10.3390/en15165884.

[63] S. S. Bhakre, P. D. Sawarkar, V. R. Kalamkar, Evaporative cooling of photovoltaic panels in dry and hot climatic conditions, Energy Sources, Part A: Recovery, Utilization, and Environmental Effects 45 (2023) 11259–11277. URL: https://doi.org/10.1080/15567036.2023.2256687. doi:10.1080/15567036.2023.2256687. arXiv:https://doi.org/10.1080/15567036.2023.2256687.

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